# Chapter 1 Quantum many-body framework

This chapter is dedicated to introducing the necessary tools needed to perform realistic material calculations using dynamical mean-field theory (DMFT) or the dynamical vertex approximation (DΓA). We start with the most central objects — the Green's functions — and subsequently assemble a complete framework which enables us to calculate a number of physical properties, e.g. the susceptibility, optical conductivity, superconductivity, pseudogap formation and more, that stem from the correlated interplay between electrons or holes.

## 1.1 One-particle Green's functions

Green's functions are fundamental tools in many-body physics, used to study the properties and behavior of interacting quantum systems. These mathematical objects encode information about the propagation of particles or excitations within a system and serve as the cornerstone for analyzing both equilibrium and nonequilibrium states. In the many-body regime, Green's functions extend beyond single-particle descriptions to include complex interactions among multiple particles. They provide insights into phenomena such as quasiparticle lifetimes, collective excitations and response functions. Specifically, one- and two-particle Green's functions describe the propagation of a single particle and the correlated motion of particle pairs, respectively. Alongside of the mathematical description of Green's function-based expressions, we also provide a pictorial description of the equations using Feynman diagrams. These Feynman diagrams allow us to easily describe interaction processes visually. Lines and vertices within the diagrams indicate particle propagators and interaction vertices, respectively. External legs will be colored gray while inner legs and interaction lines will be colored black.
In condensed matter physics, we are usually interested in finite-temperature effects. Hence, it is practical to use the so-called Matsubara formalism [1] in imaginary time by performing a Wick rotation $t \to -i\tau$ [2].

After this short introduction, let us dive in by defining the most central quantity, the two-point (one-particle) Green's function, in a system in thermal equilibrium as

$$G^{\mathbf{k}}_{12} = -\left\langle \mathcal{T}\left[\hat{c}_{\mathbf{k};1}\hat{c}^{\dagger}_{\mathbf{k};2}\right]\right\rangle, \tag{1.1}$$

where $\hat{c}_{\mathbf{k};i}$ ($\hat{c}^{\dagger}_{\mathbf{k};i}$) are fermionic annihilation (creation) operators which annihilate (create) an electron with parameters $i = \{\sigma_i, o_i, \tau_i\}$ and momentum $\mathbf{k}$ in the system. $\langle\cdot\rangle = \frac{1}{Z}\mathrm{Tr}\{e^{-\beta\hat{\mathcal{H}}}\ \cdot\}$ with $Z = \mathrm{Tr}\{e^{-\beta\hat{\mathcal{H}}}\}$ describes the thermal expectation value. Note, that the time evolution operator $e^{i\hat{\mathcal{H}}t} = e^{\hat{\mathcal{H}}\tau}$ is real after performing the Wick rotation to imaginary time and the Boltzmann weight $e^{\beta\hat{\mathcal{H}}}$ describes nothing more than an additional evolution in imaginary time. Lastly, $\mathcal{T}[\cdot]$ is the imaginary time ordering operator, where

$$\mathcal{T}\left[\hat{c}_1^{(\dagger)}(\tau_1)\hat{c}_2^{(\dagger)}(\tau_2)\right] = \Theta(\tau_1-\tau_2)\hat{c}_1^{(\dagger)}(\tau_1)\hat{c}_2^{(\dagger)}(\tau_2) - \Theta(\tau_2-\tau_1)\hat{c}_2^{(\dagger)}(\tau_2)\hat{c}_1^{(\dagger)}(\tau_1). \tag{1.2}$$

In simple quantum-mechanical terms, Green's functions are often introduced as probability amplitudes describing the propagation of a particle from one point in space and time to another. While this picture provides useful intuition, it becomes too restrictive in our context. Here, Green's functions are defined as thermal averages, i.e. they involve a summation over many states, which obscures a direct interpretation as probability amplitudes. Instead, their physical significance is more naturally understood in terms of response functions: Green's functions are directly related to measurable quantities such as spectral functions, densities of states, susceptibilities and conductivities. They are the most central objects in our theory and enter all Feynman diagrams; hence they read

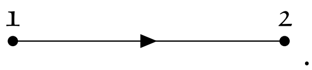
$G_{12} =$ 


**Figure 1.1** – Diagrammatic representation of the one-particle Green's function $G_{12}$. The electron is annihilated at 1, propagates in imaginary time from $\tau_1$ to $\tau_2$ and is created at 2.

In homogeneous, i.e. translationally invariant, systems the Green's function reflects this symmetry and becomes invariant under spatial translations. This means that $G_{12}(\mathbf{R}_1,\mathbf{R}_2) = G_{12}(\mathbf{R}_1-\mathbf{R}_2,\mathbf{0}) = G_{12}(\mathbf{R})$. In Fourier space, due to momentum conservation at points 1 and 2, the Green's function is only dependent on a single momentum parameter, $G_{12}(\mathbf{k}_1,\mathbf{k}_2) = G_{12}(\mathbf{k}_1-\mathbf{k}_2,\mathbf{0}) = G_{12}(\mathbf{k})$. Furthermore, if the system possesses time-translational invariance or is stationary, i.e. the Hamiltonian is not explicitly dependent on imaginary time, then $G_{12}(\tau_1,\tau_2) = G_{12}(\tau_1-\tau_2,0) = G_{12}(\tau)$. If the system additionally possesses SU(2) symmetry, the Green's functions are spin-diagonal ($G_{\uparrow\downarrow} = G_{\downarrow\uparrow} = 0$) and the diagonal components are identical $G_{\uparrow\uparrow} = G_{\downarrow\downarrow} = G$. This also holds for the self-energy, which will be discussed later. Additionally, the *fermionic* Green's function can be shown to be anti-periodic in imaginary time with a period of $\beta$[1],

$$G_{12}(\tau) = -G_{12}(\beta-\tau) \quad \text{for } \tau > 0 \qquad \text{and} \tag{1.3a}$$

$$G_{12}(\tau) = -G_{12}(\beta+\tau) \quad \text{for } \tau < 0. \tag{1.3b}$$

This anti-periodicity in time leads to the Fourier transform to be defined over a discrete set of imaginary frequencies, the the so-called Matsubara frequencies, which for fermionic quantities are given by $\nu_n = (2n+1)\frac{\pi}{\beta}$. In contrast, *bosonic* Green's functions are periodic in imaginary time with period $\beta$ and therefore the Fourier transform includes the set of bosonic Matsubara frequencies $\omega_n = (2n)\frac{\pi}{\beta}$. In the following, we will always denote fermionic Matsubara frequencies with $\nu_n$ and bosonic ones with $\omega_n$. The fermionic- and bosonic-like Fourier transforms of $G_{12}(\tau)$ become, respectively,

$$G_{12}(i\nu_n) = \int_0^\beta d\tau\, G_{12}(\tau)e^{i\nu_n\tau} \qquad \text{and} \qquad G_{12}(i\omega_n) = \int_0^\beta d\tau\, G_{12}(\tau)e^{i\omega_n\tau} \tag{1.4}$$

[1] This originates from the Boltzmann term $e^{\beta\hat{\mathcal{H}}}$ in the thermal expectation value and restricts the imaginary time domain to $\tau \in [-\beta,\beta)$.

and only require the knowledge of Green's functions for positive imaginary times $\tau$. The inverse Fourier transforms are given by

$$G_{12}(\tau) = \frac{1}{\beta}\sum_n G_{12}(i\nu_n)e^{-i\nu_n\tau} \qquad \text{and} \qquad G_{12}(\tau) = \frac{1}{\beta}\sum_n G_{12}(i\omega_n)e^{-i\omega_n\tau}. \tag{1.5}$$

By expressing the fermionic Green's function in Eq. (**??**) in its spectral representation, performing the discrete Fourier transform and subsequently transforming to Matsubara frequencies, we obtain

$$G_{12}(i\nu_n) = \frac{1}{Z}\sum_{mn}\left(e^{-\beta E_n} + e^{-\beta E_m}\right)\frac{\langle n|\hat{c}_1|m\rangle\langle m|\hat{c}_2^\dagger|n\rangle}{i\nu - (E_m - E_n)}. \tag{1.6}$$

If we take $\mathbf{R}_1 = \mathbf{R}_2$, which corresponds to the *local* Green's function, we can introduce the **k**-resolved spectral function

$$A(\mathbf{k},\nu) = \frac{1}{Z}\sum_{mn} e^{-\beta E_n}\left(1 + e^{-\beta\nu}\right)|\langle n|\hat{c}_{\mathbf{k}}|m\rangle|^2\delta(\nu - E_m - E_n), \tag{1.7}$$

where

$$G(\mathbf{k}, i\nu_n) = \int_{\mathbb{R}} d\nu\,\frac{A(\mathbf{k},\nu)}{i\nu_n - \nu}. \tag{1.8}$$

The spectral function $A(\mathbf{k},\nu)$ characterizes the probability of adding or removing a particle with momentum **k** and spin $\sigma$ at energy $\nu$. In this sense, it directly reflects the processes measured in angle-resolved photoemission spectroscopy (ARPES) and inverse photoemission, where an electron is removed from or injected into the system. For this reason, $A(\mathbf{k},\nu)$ is referred to as the single-particle (**k**-resolved) spectral function. Moreover, since it represents a normalized probability distribution in energy, the integral of $A(\mathbf{k},\nu)$ over all frequencies equals unity. It is directly measurable using techniques like angle-resolved photoemission spectroscopy (ARPES) [3, 4], which probes electronic states in the material and reveals electronic structure properties, such as band gaps and quasiparticle dispersions. Peaks in $A(\mathbf{k},\nu)$ correspond to quasiparticle states, with their position indicating the energy of these states and their width related to the lifetime or decay rate of the quasiparticles. A narrow peak indicates a well-defined quasiparticle with a long lifetime and weak decay rate, while a broad peak suggests the contrary. The spectral function is a key quantity in many-body theory as it connects real-world experimental results to the Green's function formalism, as is shown in Refs. [3, 4] and the following. By performing analytic continuation in the upper complex plane ($i\nu \to \nu + i0^+$), one can define the retarded Green's function as

$$G^R(\mathbf{k},\nu) = \int_{\mathbb{R}} d\nu'\,\frac{A(\mathbf{k},\nu')}{\nu - \nu' + i0^+}, \tag{1.9}$$

from where one obtains through the Kramers-Kronig relations

$$A(\mathbf{k},\nu) = -\frac{1}{\pi}\Im G^R(\mathbf{k},\nu). \tag{1.10}$$

Thus, the spectral function is directly connected to the imaginary part of the retarded Green's function, connecting many-body theory with physical experiments.

As an example, which will prove itself useful for the future, let us consider non-interacting electrons in a system with translational symmetry that are described by the Hamiltonian

$$\hat{\mathcal{H}}_0 = \sum_{\mathbf{k};12} \varepsilon_{12}(\mathbf{k})\hat{c}^\dagger_{\mathbf{k};1}\hat{c}_{\mathbf{k};2}, \tag{1.11}$$

where $\varepsilon_{12}(\mathbf{k}) = -\sum_{\mathbf{R};12} t_{12}(\mathbf{R})e^{i\mathbf{k}\mathbf{R}}$ is the momentum dispersion of the band and is measured with respect to the chemical potential $\mu$. In this case, it is easy to show, that

$$G_{0;12}(\nu,\mathbf{k}) = [i\nu\delta_{12} - \varepsilon_{12}(\mathbf{k})]^{-1}. \tag{1.12}$$

For the one-particle spectral function, we find in this case

$$A_{12}(\varepsilon,\mathbf{k}) = \delta(\varepsilon - \varepsilon_{\mathbf{k};12}), \tag{1.13}$$

corresponding to stable quasiparticles with infinite lifetime, due to zero width of the quasiparticle peak of $A_{12}(\varepsilon,\mathbf{k})$.

A direct computation of the Green's function as defined in Eq. (**??**) generally incurs an exponential increase in computational cost with lattice size. While this approach can be carried out exactly for a limited class of Hamiltonians, it becomes infeasible for the Hubbard Hamiltonian, which we employ throughout this work. A standard approach, since the non-interacting situation can be treated straightforwardly, is to begin with the non-interacting case and construct a perturbative expansion in terms of the interaction around it. In this context, we will start with the non-interacting case described above. The perturbation expansion in the interaction $U$ will not be derived in detail in this thesis, as there are many wonderful resources that explain it very thoroughly [5, 6], thus we will only sketch it here. In essence, the expansion is constructed using the so-called S-matrix and uses Wick's theorem [2, 7, 8] and the linked cluster theorem [9]. The expansion begins by specifying the term to be expanded, for instance $\hat{\mathcal{H}}_I$ of Eq. (**??**) in the case of an interaction expansion:

$$\hat{\mathcal{H}}_I = \frac{1}{2}\sum_{1234} U_{1234}\hat{c}^\dagger_1\hat{c}^\dagger_3\hat{c}_4\hat{c}_2. \tag{1.14}$$

The expansion of the Green's function then reads

$$G_{12}(\tau) = -\sum_{n=0}^{\infty}\frac{(-1)^n}{n!}\int_0^\beta d\tau_1 \cdots \int_0^\beta d\tau_n \left\langle \mathcal{T}\left[\hat{c}_1\hat{c}^\dagger_2\hat{\mathcal{H}}_I(\tau_1)\cdots\hat{\mathcal{H}}_I(\tau_n)\right]\right\rangle_0^{\text{conn}}, \tag{1.15}$$

where $\langle\cdot\rangle_0^{\text{conn}}$ now indicates that only connected diagrams are considered in the expansion[2]. We can transform this into a diagrammatic representation by applying Wick contractions, which allow us to collect pairs of creation and annihilation operators to form independent expectation values. In the case of the $n = 1$ term in the perturbation expansion of the Green's function,

$$\begin{aligned} G_{1;12}(\tau) = \sum_{abcd}\int_0^\beta d\tau_1\, U_{abcd}\times \\ \times\Big[ -\left\langle\mathcal{T}\left[\hat{c}_1(\tau)\hat{c}^\dagger_a(\tau_1)\right]\right\rangle_0 \left\langle\mathcal{T}\left[\hat{c}_d(\tau_1)\hat{c}^\dagger_c(\tau_1)\right]\right\rangle_0 \left\langle\mathcal{T}\left[\hat{c}_b(\tau_1)\hat{c}^\dagger_2(0)\right]\right\rangle_0 \\ + \left\langle\mathcal{T}\left[\hat{c}_1(\tau)\hat{c}^\dagger_a(\tau_1)\right]\right\rangle_0 \left\langle\mathcal{T}\left[\hat{c}_b(\tau_1)\hat{c}^\dagger_c(\tau_1)\right]\right\rangle_0 \left\langle\mathcal{T}\left[\hat{c}_d(\tau_1)\hat{c}^\dagger_2(0)\right]\right\rangle_0 \Big]. \end{aligned} \tag{1.16}$$

The remaining expectation values are nothing more than minus the non-interacting Green's function as seen from Eq. (**??**). The diagrammatic representation of the first order term is shown in Fig. **??**.

[2] Disconnected diagrams are those that consist of two or more separate, unlinked parts.

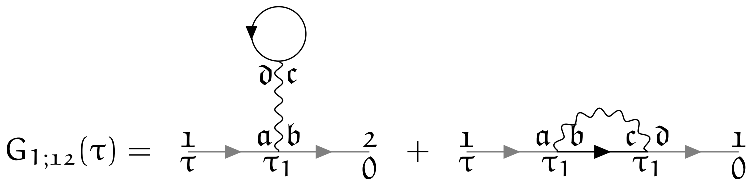



**FIGURE 1.2** – First order diagrams for the single-particle Green's function. The corresponding diagrams are coined the Hartree- and the Fock-term, respectively. As the name suggests, taking only these two diagrams as the whole perturbation expansion leads to the Hartree-Fock approximation, formulated as a diagrammatic theory.

## 1.2 Self-energy

The one-particle Green's function perturbative expansion is already much simplified in the graphical portrayal. However, the number of diagrams increases exponentially with order $n$[3]. Therefore, the Feynman diagram approach would be of little practical use if the only way to progress was to calculate each diagram one at a time. However, in reality the technique is actually quite useful, since it hints on how to sum up the perturbative series. Let us therefore review how it applies to single-particle Green's functions. First, let us define another compound index which will further levitate readability. We will set $k = \{\nu, \mathbf{k}\}$ and $q = \{\omega, \mathbf{q}\}$ as a compound index including momentum and frequency in one index for fermionic (k) and bosonic (q) frequencies and momenta. Note, that if we write $\nu$ instead of k, we refer to the *local* part of the quantity with no momentum dependence instead of the full *non-local* one.
Single-particle diagrams in general can be split up into two classes: (i) diagrams that, by cutting a single internal Green's function line, transform into two lower-order diagrams, which are called one-particle reducible and (ii) diagrams that are not one-particle reducible, which are called one-particle irreducible (1PI). Let us define as the one-particle self-energy $\Sigma^k_{12}$ the sum of all irreducible diagrams without external legs. For example, the black part of the diagrams in Fig. **??** corresponds to the first-order diagrams of the self-energy. It is then straightforward to rewrite Eq. (**??**) in terms of the self-energy, yielding

$$G^k_{12} = G^k_{0,12} + \sum_{ab} G^k_{0,1a}\Sigma^k_{ab}G^k_{b2}, \tag{1.17}$$

which results in - when taking Eq. (**??**) as the noninteracting Green's function and inverting in frequency and momentum space - a way to express the full (interacting) single-particle Green's function in terms of the self-energy

$$G^k_{12} = \left[i\nu_n\delta_{12} - \varepsilon_{12}(\mathbf{k}) - \Sigma^k_{12}\right]^{-1}. \tag{1.18}$$

Eq. (**??**) is commonly known as the Dyson equation for the single-particle Green's function and for completeness, its diagrammatic representation can be found in Fig. **??**. It is a geometric series, where each term in the sum subsequently adds irreducible diagrams that are pieced together by non-interacting Green's functions to generate all possible diagrams.

[3] For $n = 2$ there exist 10 diagrams, for $n = 7$ the number of diagrams is well over a million.

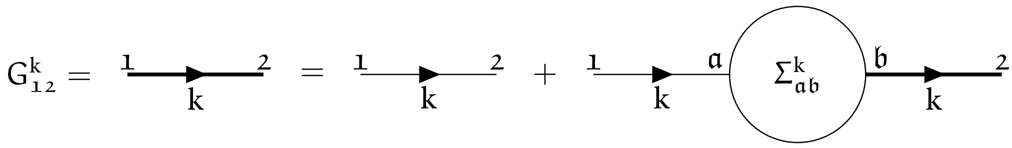



**FIGURE 1.3** – Diagrammatic representation of Eq. (??). The full Green's function $G^k_{12}$ is drawn as a thick black line.

The self-energy can be obtained by simply inverting Eq. (??),

$$\Sigma^k_{12} = (G^k_{0;12})^{-1} - (G^k_{12})^{-1}. \tag{1.19}$$

Thus, perturbation theory can be done in two distinct ways: the simpler approach is to determine the Green's function directly, for example, up to order $n$, the second method is to compute the self-energy up to order $n$ and insert its expression in the Dyson equation Eq. (??). This gives an approximate Green's function containing each order of perturbation theory with only a subset of all possible diagrams. As a result, Eq. (??) represents a way to sum up perturbation theory and is actually more relevant from a physical standpoint. Indeed, many methods leverage the usefulness of this approach, for example the dynamical mean-field theory or the dynamical vertex approximation, as we will see later.
Let us add some physical background to the self-energy in the following. $\Sigma$ represents the effects of the interaction on the single electron propagators. The real and imaginary part of the self-energy have significant impact on the physical properties of the system. The real part of the self-energy contributes to an energy shift of the electronic states of the single-particle energy levels $\varepsilon(\mathbf{k})$ due to interactions. In contrast, the imaginary part of the self-energy is related to the decay or lifetime of quasiparticles and introduces a spectral broadening or damping of the electronic states, indicating how long an excitation or quasiparticle persists before interacting with other excitations or scattering events. A larger imaginary part implies a shorter lifetime and stronger interactions. The first order derivative in frequency, $\frac{\partial \Sigma^k}{\partial \nu}$, contributes to an effective renormalization of the bare electron mass. It also leads to a redistribution of spectral weight in the one-particle spectral function, which provides information about the distribution of energy levels available for excitations. A broader spectral function, whose broadening is controlled by the imaginary part of the self-energy, typically indicates stronger interactions and a more correlated system.

## 1.3 Two-particle Green's functions

In analogy to the one-particle Green's function, the two-particle Green's function describes the amplitude of the propagation of two particles, two holes or a particle and a hole. A similar expression to Eq. (??) can be formulated for the two-particle Green's function

$$G^{qkk'}_{1234} = -\left\langle \mathcal{T}\left[\hat{c}_{\mathbf{k};1}\hat{c}^\dagger_{\mathbf{k}-\mathbf{q};2}\hat{c}_{\mathbf{k}'-\mathbf{q};3}\hat{c}^\dagger_{\mathbf{k}';4}\right]\right\rangle. \tag{1.20}$$

It describes two particles being created in the system at times $\tau_2$ and $\tau_4$, propagating in the system until they are removed again at $\tau_1$ and $\tau_3$, respectively. Assuming a stationary Hamiltonian that satisfies time-translation invariance, it is possible to eliminate one imaginary time variable by setting it zero. Then, the Green's function only depends on three time variables and through Fourier transform, also only on three frequency variables. Furthermore, since momentum is conserved, the two-particle Green's function is parameterized only by three independent momenta instead of four. Furthermore, similar to the case for the one-particle Green's function, the absence of spin-orbit coupling reduces the number of spin degrees of freedom

$$G^{qkk'}_{\sigma\sigma';1234} = G^{qkk'}_{\sigma\sigma\sigma'\sigma';1234} \qquad \text{and} \tag{1.21a}$$
$$G^{qkk'}_{\overline{\sigma\sigma'};1234} = G^{qkk'}_{\sigma\sigma'\sigma'\sigma;1234}. \tag{1.21b}$$

In this case, the total incoming and outgoing spin must be equal and this puts a heavy constraint on the possible spin combinations,

$$\sigma_1 = \sigma_2 = \sigma_3 = \sigma_4, \qquad (\sigma_1 = \sigma_2) \neq (\sigma_3 = \sigma_4) \qquad \text{and} \qquad (\sigma_1 = \sigma_4) \neq (\sigma_2 = \sigma_3), \tag{1.22}$$

totaling to only six unique spin combinations, each of which can be seen in Fig. ??. Furthermore, if

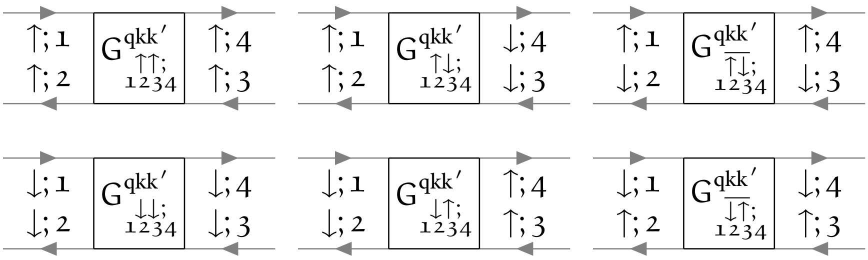



**FIGURE 1.4** – In the case of absent spin-orbit coupling, these are the only non-vanishing spin combinations of the two-particle Green's function.

one describes quantities in the paramagnetic phase, where the system is SU(2)-symmetric, there are only two independent spin configurations,

$$G^{qkk'}_{\sigma\sigma';1234} = G^{qkk'}_{(-\sigma)(-\sigma');1234} = G^{qkk'}_{\sigma'\sigma;1234} \qquad \text{and} \tag{1.23a}$$
$$G^{qkk'}_{\sigma\sigma;1234} = G^{qkk'}_{\sigma(-\sigma);1234} + G^{qkk'}_{\overline{\sigma(-\sigma)};1234}. \tag{1.23b}$$

For these two spin combinations, the (d)ensity, (m)agnetic, (s)inglet and (t)riplet channel[4] specified in the following are a particularly helpful choice of notation, as we shall see in the upcoming sections:

$$G^{qkk'}_{d;1234} = G^{qkk'}_{\uparrow\uparrow;1234} + G^{qkk'}_{\uparrow\downarrow;1234}, \tag{1.24a}$$
$$G^{qkk'}_{m;1234} = G^{qkk'}_{\uparrow\uparrow;1234} - G^{qkk'}_{\uparrow\downarrow;1234} = G^{qkk'}_{\overline{\uparrow\downarrow};1234}, \tag{1.24b}$$
$$G^{qkk'}_{s;1234} = G^{qkk'}_{\uparrow\downarrow;1234} - G^{qkk'}_{\overline{\uparrow\downarrow};1234} \qquad \text{and} \tag{1.24c}$$
$$G^{qkk'}_{t;1234} = G^{qkk'}_{\uparrow\downarrow;1234} + G^{qkk'}_{\overline{\uparrow\downarrow};1234}. \tag{1.24d}$$

Besides the spin-symmetries discussed above, the Green's function also inherits other properties of the system, such as time-reversal symmetry, which manifests itself by

$$G^{qkk'}_{1234} = G^{\bar{q}\bar{k}'\bar{k}}_{4321}, \tag{1.25}$$

[4] As we will see later, in ph/$\overline{\text{ph}}$-notation, the Bethe-Salpeter equation will become diagonal for the d/m spin combination as well as for pp-notation in the s/t spin combination.

where $\overline{k} = \{\nu, -\mathbf{k}\}$ is the time-reversed compound momentum variable. Furthermore, the two-particle Green's function satisfies the crossing symmetry, which is a direct manifestation of Pauli's principle. Crossing symmetry refers to the antisymmetric property of the Green's function with respect to the exchange of the incoming and outgoing fermions

$$G^{qkk'}_{\sigma\sigma';1234} = -G^{(k'-k)(k'-q)k'}_{\overline{\sigma'\sigma};3214} \tag{1.26a}$$

$$= -G^{(k-k')k(k-q)}_{\overline{\sigma\sigma'};1432} \tag{1.26b}$$

$$= G^{(-q)(k'-q)(k-q)}_{\sigma'\sigma;3412}. \tag{1.26c}$$

The symmetries Eq. (??) - Eq. (??) describe the frequency changes that occur by exchanging the positions of the gray fermion lines, leaving the vertex unchanged. The last line corresponds to a full swap of the incoming and outgoing particle labels. The crossing symmetry is visually shown in Fig. ??. Lastly, the Green's function also possesses symmetry with respect to complex conjugation,

$$\left(G^{qkk'}_{\sigma\sigma';1234}\right)^* = G^{(-q)(-k)(-k')}_{\sigma'\sigma;4321}. \tag{1.27}$$

At the beginning of the section it was stated that the (four-point) two-particle Green's function

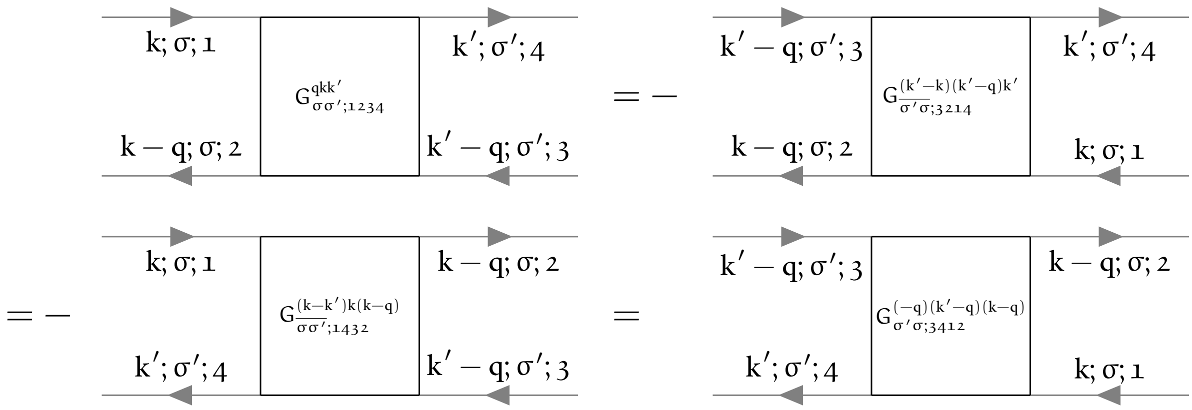



**FIGURE 1.5** – Diagrammatic representation of the crossing symmetry relations in order of Eq. (??) - Eq. (??).

only requires three frequency arguments to be fixed since the last one is automatically given by momentum conservation. There are three equivalent choices (coined channels) of independent frequencies (with $\nu$, $\nu'$ being a fermionic and $\omega$ a bosonic Matsubara frequency)

$$\text{ph-notation: } \{\nu_1 = \nu, \quad \nu_2 = \nu - \omega, \quad \nu_3 = \nu' - \omega, \quad \nu_4 = \nu'\}, \tag{1.28a}$$

$$\overline{\text{ph}}\text{-notation: } \{\nu_1 = \nu, \quad \nu_2 = \nu', \quad \nu_3 = \nu' - \omega, \quad \nu_4 = \nu - \omega\} \quad \text{and} \tag{1.28b}$$

$$\text{pp-notation: } \{\nu_1 = \nu, \quad \nu_2 = \omega - \nu', \quad \nu_3 = \omega - \nu, \quad \nu_4 = \nu'\}. \tag{1.28c}$$

To show the effects of this frequency shift and how they affect the labels of the diagrams, we show the two-particle Green's function in the three notations of Eq. (??) - Eq. (??) in Fig. ?? below. Obviously, the choice of frequency convention should not have any impact on the physical content of the Green's function, hence it is possible to switch between these channels by applying channel-specific frequency shifts [10],

$$G^{qkk'}_{ph;1234} = G^{(k-k')k(k-q)}_{\overline{ph};1234}, \tag{1.29a}$$

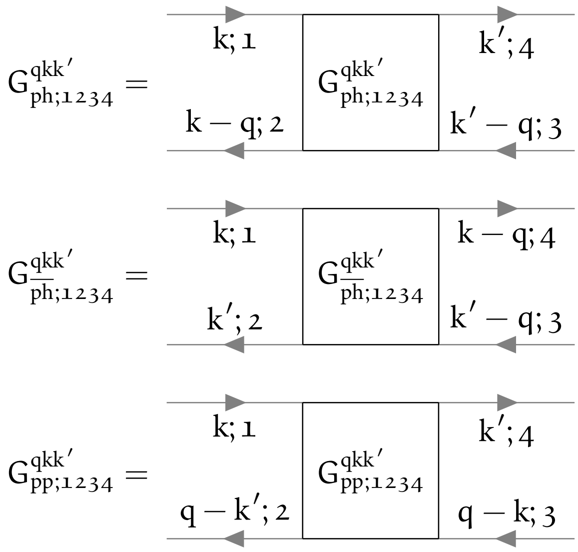



**FIGURE 1.6** – Two-particle Green's functions in the different frequency notations of Eq. (??) - Eq. (??) expressed in Feynman diagrams. The frequency notation is encoded in an additional subscript, $\{ph, \overline{ph}, pp\}$.

$$G^{qkk'}_{\overline{ph};1234} = G^{(k-k')k(k-q)}_{ph;1234}, \tag{1.29b}$$
$$G^{qkk'}_{ph;1234} = G^{(k+k'-q)kk'}_{pp;1234} \quad \text{and} \tag{1.29c}$$
$$G^{qkk'}_{pp;1234} = G^{(k+k'-q)kk'}_{ph;1234}. \tag{1.29d}$$

Let us finally note that the two-particle Green's function represents all diagrams that are possible that involve two electrons, two holes or an electron and a hole. It can therefore be split into two parts: (i) a part, where the electrons do not interact with each other and propagate independently; and (ii) a part, where the electrons do interact with each other through an infinite number of processes. (ii) is commonly called the connected two-particle Green's function and can be mathematically formulated as

$$G^{qkk'}_{\sigma\sigma';1234} = G^{conn;qkk'}_{\sigma\sigma';1234} + \delta_{q0} G^{k}_{\sigma;12} G^{k'}_{\sigma';34} - \delta_{\sigma\sigma'}\delta_{kk'} G^{k}_{\sigma;14} G^{k-q}_{\sigma;32}. \tag{1.30}$$

Diagrammatically, the connected two-particle Green's function contains all diagrams, where two propagating electrons interact with each other. The first terms up to interaction order two are given in Fig. ??. The other terms in Eq. (??) are more straightforward, they merely describe the electrons propagating in the system without ever interacting with the other. The diagrammatic content of the second and third term of Eq. (??) can be seen in Fig. ??. The connected two-particle Green's function without the external legs is commonly called the full vertex $F$, where by convention an additional minus sign is introduced,

$$G^{conn;qkk'}_{1234} = -\frac{1}{\beta} \sum_{abcd} G^{k}_{1a} G^{k-q}_{b2} F^{qkk'}_{abcd} G^{k'-q}_{3c} G^{k'}_{d4}. \tag{1.31}$$

By the previous definition, the vertex function $F^{qkk'}_{1234}$ contains all diagrams that connect two incoming and two outgoing lines. All diagrams contained in the full vertex $F^{qkk'}_{1234}$ can be classified by their two-particle reducibility (2PR). A diagram is called two-particle reducible, if it can be split into

$$G^{\mathrm{conn;qkk'}}_{\sigma\sigma';1234} =$$

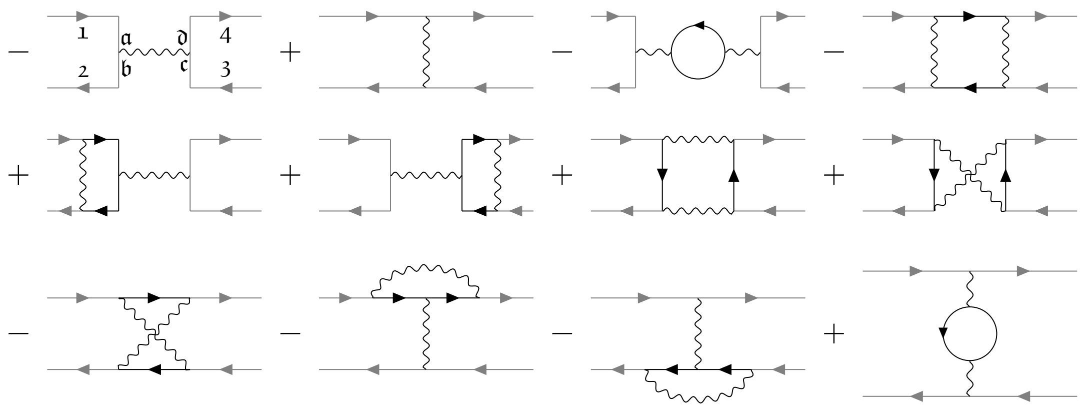


**Figure 1.7** – Diagrammatic representation of the two-particle connected Green's function up to order two. It includes the possible interaction processes that occur between two propagating electrons. The particle labels are only written down for the first diagram, the other diagrams are labeled analogously.

$$\delta_{q0} G^{k}_{\sigma;12} G^{k'}_{\sigma';34} - \delta_{\sigma\sigma'}\delta_{kk'} G^{k}_{\sigma;14} G^{k-q}_{\sigma;32} =$$

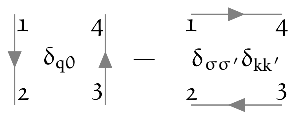


**Figure 1.8** – Disconnected part (second and third term on the right hand side of Eq. (??)) of the two-particle Green's function.

two separate parts by cutting two internal Green's function lines. Furthermore, this classification is described by three channels, $\{ph, \overline{ph}, pp\}$[5], depending on which incoming and outgoing lines are being separated by this "cutting"-process[6]. The ph-channel corresponds to those diagrams, where the sub-parts connect 12 and 34, for the $\overline{ph}$-channel 14 and 23 and for the pp-channel 13 and 24 are connected. These reducible diagrams are categorized in $\Phi^{qkk'}_{ph;1234}$, $\Phi^{qkk'}_{\overline{ph};1234}$ and $\Phi^{qkk'}_{pp;1234}$, respectively. All diagrams that are not two-particle reducible are grouped together in the quantity $\Lambda^{qkk'}_{1234}$. This procedure is called the parquet decomposition and reads mathematically

$$F^{qkk'}_{1234} = \Lambda^{qkk'}_{1234} + \Phi^{qkk'}_{ph;1234} + \Phi^{qkk'}_{\overline{ph};1234} + \Phi^{qkk'}_{pp;1234}. \tag{1.32}$$

Another possibility is to split up the full vertex $F$ in reducible and irreducible diagrams in a specific channel $r \in \{ph, \overline{ph}, pp\}$,

$$F^{qkk'}_{1234} = \Gamma^{qkk'}_{r;1234} + \Phi^{qkk'}_{r;1234}, \tag{1.33}$$

[5]The attentive reader remembers that there are also three (identically-named) frequency notations that were introduced previously. The channel reducibility and frequency notation are — in general — detached. From now on, if only one subscript $r \in \{ph, \overline{ph}, pp\}$ is given per vertex quantity, it means that the channel reducibility will belong to $r$ and the frequency notation will be ph-like. Otherwise, the channel reducibility and frequency notation will be denoted in the sub- and superscript of the variable, respectively. This will be relevant later when discussing the Bethe-Salpeter equation (BSE), see Sec. ??. The BSE is only diagonal in the bosonic transfer momentum and frequency $q$ if the channel reducibility is the same as the frequency notation.

[6]A given diagram is reducible in *at most one* of these three channels, hence making it characterized fully by the two-particle reducibility and we obtain Eq. (??).

where $\Gamma^{qkk'}_{r;1234}$ is the irreducible vertex in channel r and $\Phi^{qkk'}_{r;1234}$ the corresponding reducible one. Note, that in channel r irreducible diagrams might still be reducible in another channel $r' \neq r$. For example, $\Gamma^{qkk'}_{\overline{ph};1234}$ contains all fully irreducible diagrams, but also all diagrams which are reducible in channels ph and pp,

$$\Gamma^{qkk'}_{ph;1234} = \Lambda^{qkk'}_{1234} + \Phi^{qkk'}_{\overline{ph};1234} + \Phi^{qkk'}_{pp;1234}. \tag{1.34}$$

This will be important later in Sec. ??, where we will discuss the Bethe-Salpeter equations. Furthermore, the following relations for the ph and $\overline{ph}$ channel for $\Phi^{qkk'}_{r;1234}$ hold:

$$\Phi^{qkk'}_{\overline{ph};1234;\sigma\sigma'} = -\Phi^{(k'-k)(k'-q)k'}_{ph;3214;\overline{\sigma'\sigma}}, \tag{1.35a}$$

$$= -\Phi^{(k-k')k(k-q)}_{ph;1432;\overline{\sigma'\sigma}}. \tag{1.35b}$$

From this one can construct the density and magnetic contributions of $\Phi^{qkk'}_{\overline{ph};1234}$ in terms of $\Phi^{qkk'}_{ph;1234}$

$$\Phi^{qkk'}_{d;\overline{ph};1234} = -\frac{1}{2}\Phi^{(k'-k)(k'-q)k'}_{d;ph;3214} - \frac{3}{2}\Phi^{(k'-k)(k'-q)k'}_{m;ph;3214} \tag{1.36a}$$

$$\Phi^{qkk'}_{m;\overline{ph};1234} = -\frac{1}{2}\Phi^{(k'-k)(k'-q)k'}_{d;ph;3214} + \frac{1}{2}\Phi^{(k'-k)(k'-q)k'}_{m;ph;3214}. \tag{1.36b}$$

The relations Eq. (??) and Eq. (??) also apply to the full- and irreducible vertices $F^{qkk'}_{\overline{ph};1234;\sigma\sigma'}$ and $\Gamma^{qkk'}_{\overline{ph};1234;\sigma\sigma'}$ in a similar fashion. Note that there is no connection between the pp channel and the other two channels, hence it is not possible to transform the channel reducibility from either ph or $\overline{ph}$ to pp. The pp-channel fulfills a crossing symmetry on its own,

$$\Phi^{qkk'}_{pp;1234;\sigma\sigma'} = -\Phi^{(k-k')k(k-q)}_{pp;1432;\overline{\sigma\sigma'}}. \tag{1.37}$$

In the next section, we will briefly introduce the concept of linear response and the central physical property describing the response of a system with respect to an external perturbation[7], the susceptibility.

## 1.4 Susceptibility

Experimentally, it is not possible to directly extract information (just by "looking") about the interactions between electrons in a system. Instead, in spectroscopic experiments, a perturbation is applied to the system and one measures the system's response. In particular, one measures how expectation values $\langle\cdot\rangle$ of operators (take the multi-dimensional operator $\hat{O}_i(\tau)$ as an example) changes when an external field $h_j(\tau)$ is applied. In linear response theory, this response is — as the name suggests — taken to be linear in the perturbing field, requiring this field to be sufficiently small in order for this description to be a good approximation [11, 12],

$$\left\langle \hat{O}_i(\tau) \right\rangle_{\mathbf{h}} - \left\langle \hat{O}_i(\tau) \right\rangle_{\mathbf{h}=0} = \int_0^\beta d\tau' \chi_{ij}(\tau-\tau') h_j(\tau') + \mathcal{O}(\mathbf{h}^2), \tag{1.38}$$

[7] For example an external electromagnetic field.

where $\chi_{ij}(\tau-\tau')$ is coined physical susceptibility and does not depend on the field $h_j(\tau)$. $\chi(\tau-\tau')$ fulfills the Kramers-Kronig relations, is causal ($\chi_{ij}(\tau-\tau') = \Theta(\tau-\tau')\chi_{ij}(\tau-\tau')$) and is finite ($\forall\,\tau,\tau' :\ \exists\, C \in \mathbb{R} :\ |\chi_{ij}(\tau-\tau')| < C$). The susceptibility can be computed straightforwardly by taking the functional derivative of $\hat{O}_i(\tau)$ with respect to the field $h_j(\tau)$,

$$\chi_{ij}(\tau-\tau') = \left.\frac{\delta\left\langle \hat{O}_i(\tau')\right\rangle}{\delta h_j(\tau)}\right|_{\mathbf{h}=0} = \left\langle \mathcal{T}\hat{O}_i(\tau)\hat{O}_j(\tau')\right\rangle - \left\langle \hat{O}_i\right\rangle\left\langle \hat{O}_j\right\rangle. \tag{1.39}$$

On the two-particle level, only the density's response to variations in the one-particle energy, $\chi_d$, and the magnetization's response to variation in the electromagnetic field in the same direction, $\chi_m$, are non-zero [11],

$$\chi_d(\tau) = \chi_{nn}(\tau) = -\left.\frac{\delta\left\langle \hat{n}\right\rangle}{\delta\mu(\tau)}\right|_{\mu=0} \quad \text{and} \tag{1.40a}$$

$$\chi_m(\tau) = \chi_{ii}(\tau) = \left.\frac{\delta\left\langle \hat{\sigma}_i\right\rangle}{\delta h_i(\tau)}\right|_{\mathbf{h}=0}, \tag{1.40b}$$

where $i \in \{x, y, z\}$. The physical susceptibilities are included in the two-particle Green's function, as we will see shortly. In fact, the sum of terms corresponding to the two disconnected, horizontal Green's function lines and the vertex part is denoted as the generalized susceptibility and can be expressed as

$$\begin{aligned}\chi^{ph;qkk'}_{1234} &= \beta\left(G^{ph;qkk'}_{1234} - \delta_{q0}G^{k}_{14}G^{k'}_{32}\right)\\ &= -\beta\delta_{kk'}G^{k}_{14}G^{k-q}_{32} - \sum_{abcd} G^{k}_{1a}G^{k-q}_{b2}F^{ph;qkk'}_{abcd}G^{k'-q}_{3c}G^{k'}_{d4}\\ &= \chi^{qkk'}_{0;1234} - \frac{1}{\beta^2}\sum_{abcd}\chi^{qkk}_{0;12ba}F^{ph;qkk'}_{abcd}\chi^{qk'k'}_{0;dc34},\end{aligned} \tag{1.41}$$

where

$$\chi^{qkk'}_{0;1234} = -\beta\delta_{kk'}G^{k}_{14}G^{k-q}_{32} \tag{1.42}$$

is commonly denoted as the generalized bubble susceptibility[8]. From this one can obtain physical susceptibilities by contracting the inner legs, i.e.

$$\chi^{q}_{14} = \sum_{kk';23}\chi^{qkk'}_{1234}. \tag{1.44}$$

The diagrammatic content of the physical susceptibility is shown in Fig. **??** below. Similarly, the contracted bubble susceptibility can be written as

$$\chi^{q}_{0;14} = \sum_{kk';23}\chi^{qkk'}_{0;1234}. \tag{1.45}$$

[8] The generalized susceptibility in the particle-particle channel reads

$$\chi^{pp;qkk'}_{1234;\sigma\sigma'} = \beta\left(1-\delta_{\sigma\sigma'}\right)\left[G^{pp;qkk'}_{1234;\sigma\sigma'} - \delta_{q0}G^{-k}_{14}G^{k'}_{32}\right]. \tag{1.43}$$

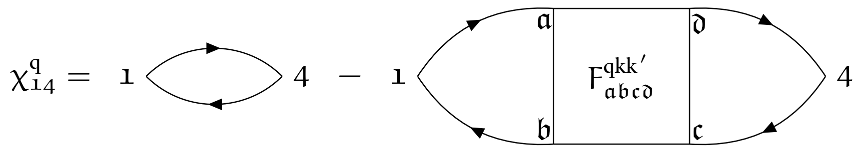



**FIGURE 1.9** – The physical susceptibility is defined as the generalized susceptibility with contracted legs, see Eq. (??).

Analogously to Eq. (??) and Eq. (??), one can write down density and magnetic spin combinations of the physical susceptibility,

$$\chi^q_d = \chi^q_{\uparrow\uparrow} + \chi^q_{\uparrow\downarrow} \qquad \text{and} \tag{1.46a}$$

$$\chi^q_m = \chi^q_{\uparrow\uparrow} - \chi^q_{\uparrow\downarrow} = \chi^q_{\overline{\uparrow\downarrow}}. \tag{1.46b}$$

Note that expressions Eq. (??) and Eq. (??) are nothing more than the Fourier-transformed versions of Eq. (??) and Eq. (??), respectively.

## 1.5 Bethe-Salpeter equation

In the section above we have specified $\Gamma^{qkk'}_{r;1234}$ as the fully irreducible vertex in channel r and $\Phi^{qkk'}_{r;1234}$ as the reducible diagrams in channel r. We can now construct the full set of $\Phi^{qkk'}_{r;1234}$ from the irreducible vertex by chaining them together with pairs of Green's functions connecting two irreducible vertices. This results in so-called ladders,

$$\Phi^{qkk'}_{r;1234} = \sum_{k_1;\mathfrak{abcd}} \Gamma^{qkk_1}_{r;12\mathfrak{ab}} G^{k_1}_{\mathfrak{bc}} G^{k_1-q}_{\mathfrak{da}} \Gamma^{qk_1k'}_{r;\mathfrak{cd}34} + \sum_{\substack{k_1k_2;\mathfrak{abcd}\\ \mathfrak{efgh}}} \Gamma^{qkk_1}_{r;12\mathfrak{ab}} G^{k_1}_{\mathfrak{bc}} G^{k_1-q}_{\mathfrak{da}} \Gamma^{qk_1k_2}_{r;\mathfrak{cdef}} G^{k_2}_{\mathfrak{fg}} G^{k_2-q}_{\mathfrak{he}} \Gamma^{qk_2k'}_{r;\mathfrak{gh}34} + \dots \tag{1.47a}$$

$$= -\frac{1}{\beta} \sum_{k_1;\mathfrak{abcd}} \Gamma^{qkk_1}_{r;12\mathfrak{ab}} \chi^{qk_1k_1}_{0;\mathfrak{badc}} \Gamma^{qk_1k'}_{r;\mathfrak{cd}34} + \frac{1}{\beta^2} \sum_{\substack{k_1k_2;\mathfrak{abcd}\\ \mathfrak{efgh}}} \Gamma^{qkk_1}_{r;12\mathfrak{ab}} \chi^{qk_1k_1}_{0;\mathfrak{badc}} \Gamma^{qk_1k_2}_{r;\mathfrak{cdef}} \chi^{qk_2k_2}_{0;\mathfrak{fehg}} \Gamma^{qk_2k'}_{r;\mathfrak{gh}34} + \dots \tag{1.47b}$$

By combining Eq. (??) and Eq. (??), one can obtain the Bethe-Salpeter equation (BSE) in all three channels [13–15],

$$F^{qkk'}_{1234} = \Gamma^{qkk'}_{ph;1234} - \frac{1}{\beta} \sum_{k_1k_2;\mathfrak{abcd}} \Gamma^{qkk_1}_{ph;12\mathfrak{ba}} \chi^{qk_1k_2}_{0;\mathfrak{abcd}} F^{qk_2k'}_{ph;\mathfrak{dc}34} \tag{1.48a}$$

$$= \Gamma^{qkk'}_{ph;1234} + \frac{1}{\beta} \sum_{k_1k_2;\mathfrak{abcd}} \Gamma^{qkk_1}_{ph;1\mathfrak{ad}4} \chi^{qk_1k_2}_{0;\mathfrak{abcd}} F^{qk_2k'}_{ph;\mathfrak{b}23\mathfrak{c}} \tag{1.48b}$$

$$= \Gamma^{qkk'}_{pp;1234} - \frac{1}{2\beta} \sum_{k_1k_2;\mathfrak{abcd}} \Gamma^{qkk_1}_{pp;1\mathfrak{b}3\mathfrak{a}} \chi^{qk_1k_2}_{0;\mathfrak{abcd}} F^{qk_2k'}_{pp;\mathfrak{d}2\mathfrak{c}4}, \tag{1.48c}$$

where we have expressed the full vertex $F^{qkk'}_{1234}$ in terms of the irreducible vertex $\Gamma^{qkk'}_{r;1234}$ and the generalized (bubble) susceptibility as a Dyson-like equation. Note that for Eq. (??) we used the same channel reducibility and frequency notation for each separate line. This results in a key property of the BSE: the equation becomes diagonal in the bosonic transfer momentum and frequency q. This can be taken advantage of in numerical computations as it makes it especially easy to parallelize over q as it is independent of all other quantum numbers. If we had not chosen the channel reducibility equal to the frequency notation, then the new $q'$ would depend on the fermionic frequencies (the precise dependency is channel-specific, but in general $q' = f(q, k, k')$) and the transformed BSE would not be diagonal in $q'$.
Eq. (??) can be rewritten in terms of susceptibilities with the help of Eq. (??),

$$\chi^{qkk'}_{1234} = \chi^{qkk'}_{0;1234} - \frac{1}{\beta^2} \sum_{k_1 k_2;abcd} \chi^{qkk_1}_{0;12ba} \Gamma^{qk_1k_2}_{ph;abcd} \chi^{qk_2k'}_{ph;dc34} \tag{1.49a}$$

$$= \chi^{qkk'}_{0;1234} + \frac{1}{\beta^2} \sum_{k_1 k_2;abcd} \chi^{qkk_1}_{0;1ad4} \Gamma^{qk_1k_2}_{\overline{ph};abcd} \chi^{qk_2k'}_{\overline{ph};b23c} \tag{1.49b}$$

$$= \chi^{qkk'}_{0;1234} - \frac{1}{2\beta^2} \sum_{k_1 k_2;abcd} \chi^{qkk_1}_{0;1b3a} \Gamma^{qk_1k_2}_{pp;abcd} \left(\chi^{qk_2k'}_{pp;d2c4} + \chi^{qk_2k'}_{0;d2c4}\right). \tag{1.49c}$$

This is the most general form of the Bethe-Salpeter equations, where energy and momentum is conserved. If the system under consideration is additionally SU(2)-symmetric, then we can formulate the BSE in terms of density and magnetic spin components,

$$F^{qkk'}_{d/m;ph;1234} = \Gamma^{qkk'}_{d/m;ph;1234} - \frac{1}{\beta} \sum_{k_1 k_2;abcd} \Gamma^{qkk_1}_{d/m;ph;12ba} \chi^{qk_1k_2}_{0;abcd} F^{qk_2k'}_{d/m;ph;dc34} \quad \text{and} \tag{1.50a}$$

$$F^{qkk'}_{d/m;\overline{ph};1234} = \Gamma^{qkk'}_{d/m;\overline{ph};1234} - \frac{1}{\beta} \sum_{k_1 k_2;abcd} \Gamma^{qkk_1}_{d/m;\overline{ph};1ad4} \chi^{qk_1k_2}_{0;abcd} F^{qk_2k'}_{d/m;\overline{ph};b23c}. \tag{1.50b}$$

Likewise, the BSE decouples for other spin-symmetric combinations in for the pp-channel. These are the (s)inglet and (t)riplet channel, where

$$F^{qkk'}_{s;pp;1234} = \Gamma^{qkk'}_{s;pp;1234} + \frac{1}{2\beta} \sum_{k_1 k_2;abcd} \Gamma^{qkk_1}_{s;pp;1b3a} \chi^{qk_1k_2}_{0;abcd} F^{qk_2k'}_{s;pp;d2c4} \quad \text{and} \tag{1.51}$$

$$F^{qkk'}_{t;pp;1234} = \Gamma^{qkk'}_{t;pp;1234} - \frac{1}{2\beta} \sum_{k_1 k_2;abcd} \Gamma^{qkk_1}_{t;pp;1b3a} \chi^{qk_1k_2}_{0;abcd} F^{qk_2k'}_{t;pp;d2c4}. \tag{1.52}$$

For a visual representation of the Bethe-Salpeter equation (??), see Fig. ??.
The BSE of Eq. (??) needs to be inverted in order to be able to yield the full vertex $F^{qkk'}_{1234}$. This can be done in a smart way by viewing the multiplications of the vertex functions in the BSE as matrix multiplications written down in grouped indices. This means that for each transfer momentum q, we can generate a matrix from a vertex by grouping the left (right) two orbital indices and fermionic frequencies into a new grouped index $n = \{n_1, n_2, k\}$ ($m = \{n_4, n_3, k'\}$), such that [16]

$$\boldsymbol{F}^{q}_{d/m;nm} = F^{qkk'}_{d/m;1234}. \tag{1.53}$$

This allows for a straightforward numerical implementation of vertex products and inversions with respect to these grouped indices. Hence, this yields for the full vertex $\boldsymbol{F}^{q}_{d/m;nm}$ obtained from the

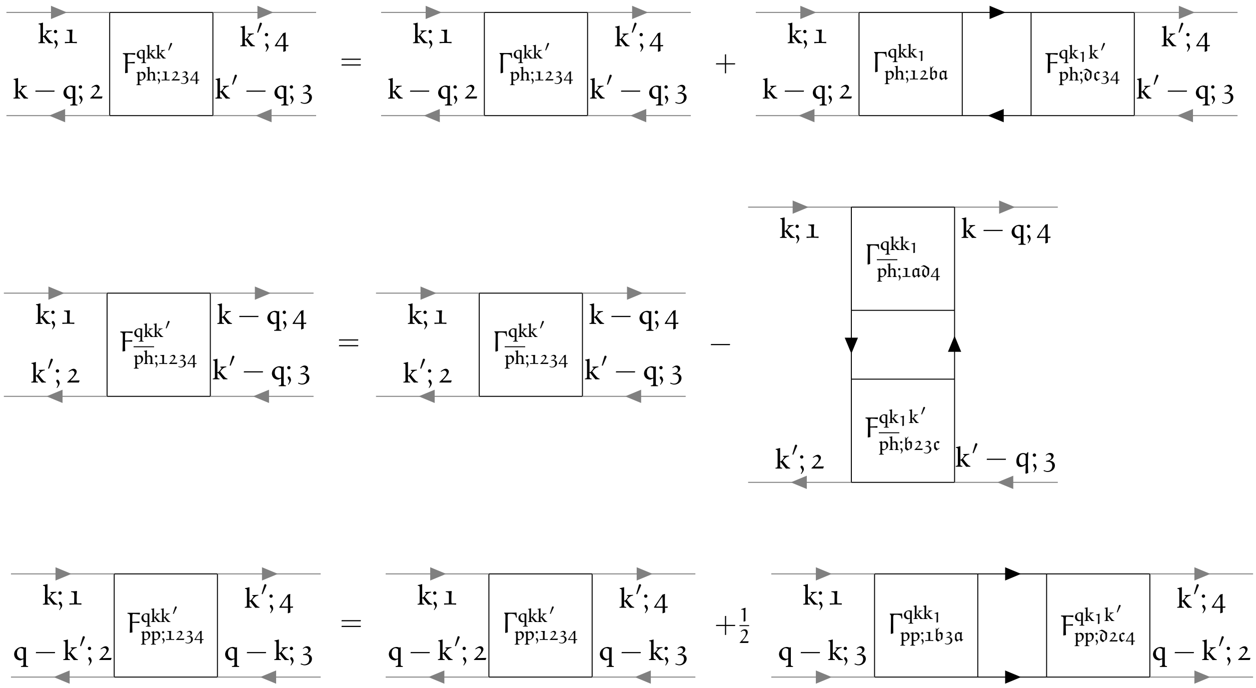


**FIGURE 1.10** – Diagrammatic version of the BSE in all three channels for arbitrary spin components, see Eq. (**??**). The lower legs are exchanged in pp-notation, hence their direction is reversed.

BSE,

$$\boldsymbol{F}^{q}_{d/m;nm} = \boldsymbol{\Gamma}^{q}_{d/m;nl}\left(\mathbb{1}_{ac} + \frac{1}{\beta}\boldsymbol{\Gamma}^{q}_{d/m;ab}\chi^{q}_{0;bc}\right)^{-1}_{lm}. \tag{1.54}$$

## 1.6 Schwinger-Dyson equation

To fill the missing gap in our (almost) complete theory of correlated electrons on a lattice, we have to consider the calculation of the self-energy which enters the many-body Green's function through Eq. (**??**). This is done by the Schwinger-Dyson equation, which is derived as an equation of motion for the self-energy [15] and connects the self-energy to the full vertex via[9] [16, 17]

$$\begin{aligned}\Sigma^{k}_{12} &= \Sigma^{k}_{HF;12} + \Sigma^{conn;k}_{12}\\ &= \underbrace{2\sum_{ab;k'}\mathcal{U}^{q=0}_{12ab}n^{k'}_{ba} - \sum_{ab,q}\mathcal{U}^{q}_{1ba2}n^{k-q}_{ba}}_{\Sigma^{k}_{HF;12}}\underbrace{-\frac{1}{\beta}\sum_{\substack{abcdef\\ qk'}}\mathcal{U}^{q}_{a1bc}\chi^{qk'k'}_{0;cbed}F^{qk'k}_{de2f}G^{k-q}_{af}}_{\Sigma^{conn;k}_{12}},\end{aligned} \tag{1.55}$$

where the first two sums in Eq. (**??**) are coined Hartree- and Fock-terms, respectively, and the third term is the vertex part and only contains connected diagrams. A visual representation of

[9] In its most general form, the self-energy $\Sigma$ also depends on the particle spin $\sigma$. In the SU(2)-symmetric case however we can omit the spin label, since $\Sigma_{\uparrow} = \Sigma_{\downarrow} = \Sigma$.

the Schwinger-Dyson equation can be seen in Fig. **??**. Here, $\mathcal{U}^{\mathbf{q}}_{1234}$ is the full non-local Coulomb interaction, written in crossing-symmetric notation [18], where we set $\mathbf{k} = \mathbf{k}' = \mathbf{0}$

$$\mathcal{U}^{\mathbf{qkk}'}_{1234;\sigma\sigma'} = \underbrace{U_{1234} + V^{\mathbf{q}}_{1234}}_{\mathcal{U}^{\mathbf{q}}_{1234}} - \delta_{\sigma\sigma'}\Big(\underbrace{U_{2314} + V^{\mathbf{k}'-\mathbf{k}}_{2314}}_{\tilde{\mathcal{U}}^{\mathbf{k}'-\mathbf{k}}_{1234}}\Big), \tag{1.56}$$

containing the purely local interaction $U$ and the purely non-local interaction $V$. Eq. (**??**) is diagrammatically displayed in Fig. **??**. The ph and $\overline{\text{ph}}$ contributions of the connected part yield separately

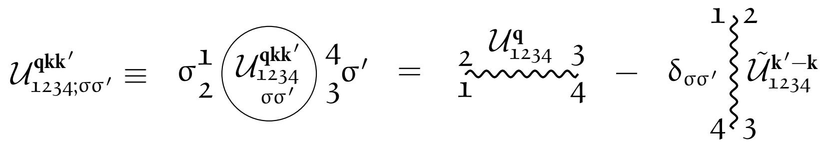



**FIGURE 1.11** – Crossing-symmetric interaction, see Eq. (**??**).

(in ph notation)[10] [16]

$$\Sigma^{\text{conn};\mathrm{k}}_{\text{ph};12} = -\frac{1}{\beta}\sum_{\substack{abcdef\\ qk'}} \mathcal{U}^{\mathbf{q}}_{a1bc}\chi^{qk'k'}_{0;cbed}F^{qk'k}_{d;de2f}G^{k-q}_{af} \qquad \text{and} \tag{1.57a}$$

$$\Sigma^{\text{conn;ph;k}}_{\overline{\text{ph}};12} = -\frac{1}{2\beta}\sum_{\substack{abcdef\\ qk'}} \tilde{\mathcal{U}}^{\mathbf{k}'-\mathbf{k}}_{a1bc}\chi^{qk'k'}_{0;cbed}\left[F^{(k'-k)(k'-q)k'}_{d;2edf} + 3F^{(k'-k)(k'-q)k'}_{m;2edf}\right]G^{k-q}_{af}. \tag{1.57b}$$

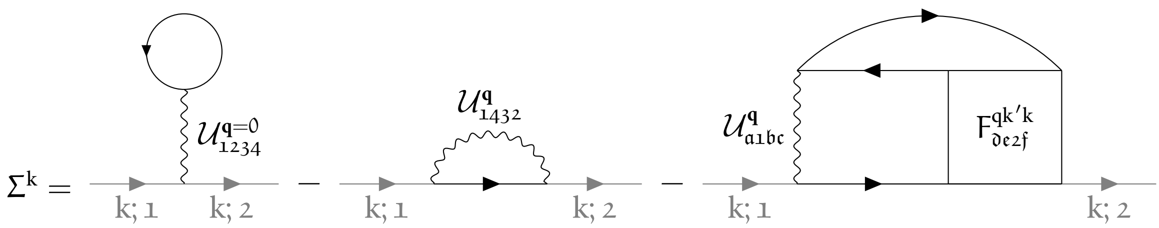



**FIGURE 1.12** – Representation of the Schwinger-Dyson equation, see Eq. (**??**).

The self-energy $\Sigma^{k}_{12}$ does not have to fulfill any crossing-symmetry relation, hence we do not use $\mathcal{U}^{\mathbf{qkk}'}_{1234;\sigma\sigma'}$ from Eq. (**??**) here but rather the two terms with $\mathcal{U}$ and $\tilde{\mathcal{U}}$ separately. The total self-energy is now the sum of the Hartree-Fock contribution and the ph/$\overline{\text{ph}}$ terms. We want to note that the Schwinger-Dyson equation can be rewritten in terms of three-leg vertices (also commonly called Hedin-vertices) instead of four-point Green's functions, see [19, 20] to eliminate the need of calculating the full vertex $F$ via an inversion of the irreducible vertex $\Gamma$[11]. This equivalent

[10] With the help of Eq. (**??**)

[11] In principle, one could retrieve the full vertex as $\boldsymbol{F}^{q}_{r} = \left[(\boldsymbol{\Gamma}^{q}_{r})^{-1} + \chi^{q}_{0}\right]^{-1}$. However, as shown in Ref. [21], the (local) irreducible vertex $\Gamma$ contains an infinite set of divergencies, hence inverting it is numerically unstable.

formulation is commonly called the Hedin form. In this form, the SDE can be rewritten in terms of susceptibilities. We start by rewriting the full vertex as a combination of susceptibilities

$$F^{qkk'}_{1234;\sigma\sigma'} = \beta^2 \left[ \left(\chi^{qkk'}_{0;1234}\right)^{-1} - \sum_{k_1 k_2;abcd} \left(\chi^{qkk_1}_{0;12ba}\right)^{-1} \chi^{qk_1k_2}_{abcd;\sigma\sigma'} \left(\chi^{qk_2k'}_{0;dc34}\right)^{-1} \right]^{-1}. \tag{1.58}$$

Here and in the following, we will restrict ourselves to the ph channel reducibility and frequency notation if not stated otherwise. Next, we define the irreducible vertex as the first-order contribution — which is the bare Coulomb interaction in crossing-symmetric notation, see Eq. (**??**) — and the rest,

$$\Gamma^{qkk'}_{1234;\sigma\sigma'} = \mathcal{U}^{\mathbf{qkk'}}_{1234;\sigma\sigma'} + \delta\Gamma^{qkk'}_{1234;\sigma\sigma'}. \tag{1.59}$$

It is then straightforward to define an auxiliary susceptibility (denoted by the asterisk $^*$)

$$\begin{aligned} \chi^{*;qkk'}_{1234;\sigma\sigma'} &= \left[ \left(\chi^{qkk}_{0;1234}\right)^{-1} + \frac{1}{\beta^2}\delta\Gamma^{qkk'}_{1234;\sigma\sigma'} \right]^{-1} \\ &= \left[ \left(\chi^{qkk}_{0;1234}\right)^{-1} + \frac{1}{\beta^2}\left(\Gamma^{qkk'}_{1234;\sigma\sigma'} - \mathcal{U}^{\mathbf{qkk'}}_{1234;\sigma\sigma'}\right) \right]^{-1}, \end{aligned} \tag{1.60}$$

with which we can rewrite the generalized susceptibility as

$$\chi^{qkk'}_{1234;\sigma\sigma'} = \chi^{*;qkk'}_{1234;\sigma\sigma'} - \sum_{\substack{k_1 k_2;abcd \\ \sigma_1\sigma_2}} \chi^{*;qkk_1}_{12ba;\sigma\sigma_1} \mathcal{U}^{\mathbf{qk_1k_2}}_{abcd;\sigma_1\sigma_2} \chi^{qk_2k'}_{dc34;\sigma_2\sigma'}. \tag{1.61}$$

Previously, the BSE was formulated as a ladder with $\Gamma^{qkk'}_{1234}$ and $\chi^{qkk}_{0;1234}$ as building blocks. The rewritten BSE can now be understood as a ladder with $\chi^{*;qkk'}_{1234;\sigma\sigma'}$ and $\mathcal{U}^{\mathbf{qkk'}}_{1234;\sigma\sigma'}$ as building blocks. Here, the auxiliary susceptibility can be seen as the irreducible part with respect to the bare Coulomb interaction, contrary to how $\Gamma^{qkk'}_{1234}$ is the set of irreducible diagrams with respect to cutting two Green function lines (or a bare susceptibility $\chi^{qkk}_{0;1234}$). Hence, this is commonly called interaction-reducibility [22]. The BSE displayed in terms of susceptibilities can be seen in Fig. **??**. If one neglects

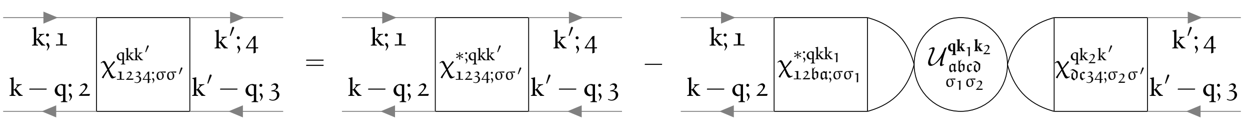



**FIGURE 1.13** – Interaction-reducible form of the Schwinger-Dyson equation (**??**) with susceptibilities.

the nonlocal $\overline{\text{ph}}$ contribution from the bare Coulomb interaction, one can rewrite the following expression in terms of sums of susceptibilities, where after some algebra,

$$\sum_{k';ab} F^{qkk'}_{12ab;\sigma\sigma'} \chi^{qk'k'}_{0;ba34} = \beta \sum_{abcd;\sigma_1} \gamma^{qk}_{12ab;\sigma\sigma'} \left( \mathbb{1}^{q}_{ba34} - \mathcal{U}^{\mathbf{q}}_{badc;\sigma\sigma_1} \chi^{q}_{cd34;\sigma_1\sigma} \right), \tag{1.62}$$

where the Fermi-Bose three-leg vertex $\gamma^{qk}_{1234;\sigma\sigma'}$ is defined as

$$\gamma^{qk}_{1234;\sigma\sigma'} = \beta \sum_{k_1;ab} \left(\chi^{qkk}_{0;12ab}\right)^{-1} \chi^{*;qkk_1}_{ba34;\sigma\sigma'}. \tag{1.63}$$

This allows us to rewrite Eq. (??) as

$$\Sigma^{k}_{12} = \Sigma^{\mathbf{k}}_{\mathrm{HF};12} - \sum_{\substack{q;abcd\\efgh}} \mathcal{U}^{\mathbf{q}}_{a1bc} \left[\gamma^{qk}_{aced}\left(\mathbb{1}_{deh2} - \mathcal{U}^{\mathbf{q}}_{defg}\chi^{q}_{gfh2}\right)\right] G^{k-q}_{bh}. \tag{1.64}$$

A visual representation of the rewritten SDE can be found in Fig. ??. This expression has some

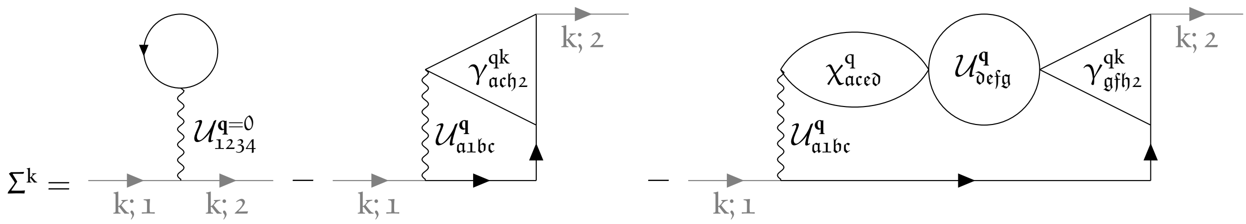



**FIGURE 1.14** – Interaction-reducible form of the Schwinger-Dyson equation (??) with three-leg vertices.

numerical advantages, since the frequency dimension of the Fermi-Bose vertices needed for the calculation of the self-energy is one smaller compared to the approach with the full vertices, hence reducing the required memory storage.

For numerical calculations with the equations in the above sections one is always restricted to a certain frequency range where the input quantities are known, which is usually limited by time and the impurity solver's capabilities. However, the full and irreducible vertex functions outside of this "core" frequency box are often not negligible[12]. For more accurate descriptions of materials, one can employ explicit asymptotics that should mimic the outer frequency structure of the vertex functions. These asymptotics can be done in many ways, from simply concatenating constant values to the region outside of the calculated one[13] to more complicated formulations using so-called kernel functions. In this thesis we will introduce two treatments of the asymptotic behavior of vertex functions: one is the kernel-function approach — starting from the pioneering work of Ref. [23] that was later developed into a full local framework [10, 24–28] — which mimics the coarse frequency structure of the vertex function outside the core frequency region; and the other is an "RPA-esque" approach [29], where a constant value outside the core frequency region of the irreducible vertex is assumed and diagrams are generated by a ladder, similar to the BSE. We will outline the two methods in detail in the next section, explain their advantages and drawbacks and reason briefly when to choose either of them. Since the former method using kernel-functions has not been generalized to non-local quantities yet at the writing of this thesis, we will derive equations that include non-local contributions to better incorporate these asymptotics in the computational frameworks that will be introduced in Ch. ??.

[12] Especially not if one is interested in low-temperature behavior. Since the frequency spacing scales with $\sim \frac{1}{\beta} \sim T$, the lower the temperature, the smaller the effective frequency box at a fixed number of frequencies. Even though the full and irreducible vertex functions decay like $\frac{1}{\nu}$ as $\nu \to \infty$, the contributions outside of the core frequency region are of importance.

[13] Like the bare Coulomb interaction $\mathcal{U}^{\mathbf{qkk}'}_{1234;\sigma\sigma'}$

## 1.7 Vertex asymptotics

A full treatment of frequency and momentum dependence of multi-orbital vertex functions severely restricts the speed and memory consumption of numerical calculations. A step towards reducing the computational cost of numerical implementations of equations involving vertex functions, such as the Bethe-Salpeter equations, see Sec. **??**, or the Schwinger-Dyson equation, see Sec. **??**, thus requires an efficient treatment of the vertex for high Matsubara frequencies. As noted above, we will describe two ways of treating of vertex asymptotics in the following.

### 1.7.1 Kernel-function asymptotics

Shortly, we will see, that the full vertex in the high-frequency regime, which generally depends on three independent frequency parameters, can be approximated well as an object with only two frequency dimensions, thus drastically reducing the storage capacity required, while at the same time also reducing the computational cost after assembling these objects. Let us start with a sketch of the basic idea of how to take advantage of certain frequency structures when handling explicit vertex asymptotics. In Fig. **??**, the frequency dependence of the local vertex functions $F^{\omega\nu\nu'}_{pp;\uparrow\downarrow}$, $\Lambda^{\omega\nu\nu'}_{pp;\uparrow\downarrow}$ and

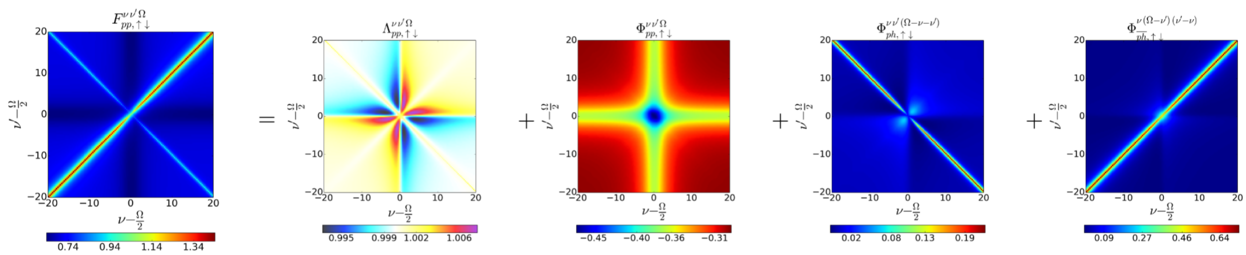



**FIGURE 1.15** – Numerical SIAM results for all local vertices contained in the parquet equation Eq. (**??**) for vanishing transfer frequency $\Omega = 0$, taken from Ref. [24]. In our convention, $\Omega$ should be replaced by $\omega$.

$\Phi^{\omega\nu\nu'}_{r;\uparrow\downarrow}$ calculated for a Single-Impurity Anderson Model (SIAM) are shown for vanishing transfer frequency $\omega$ [24]. From this, one can see three very prominent and distinct features of the full vertex $F$: (i) a constant background that differs from the (local and constant) Hubbard interaction $U$[14]; (ii) two diagonal structures, the main ($\nu = \nu'$) and secondary ($\nu = -\nu'$) diagonal; and (iii) a "cross"-like structure along $\nu = \pm\frac{\pi}{\beta}$ and $\nu' = \pm\frac{\pi}{\beta}$. While only shown for the antiparallel spin combination $\uparrow\downarrow$ and vanishing transfer frequency $\omega = 0$, the features are analogous for the parallel spin combination $\uparrow\uparrow$ and for finite $\omega \neq 0$. These three attributes do *not* decay for high Matsubara frequencies $\nu$ and $\nu'$ and therefore indicate complex asymptotic behaviour. The individual asymptotic contributions of the fully irreducible vertex $\Lambda$ and the reducible vertex in channel $r$ to the full vertex $F$ are written in detail in Ref. [24]. Summarized, $\Lambda$ adds a constant background of size $U$, $\Phi_{ph}$ adds a secondary diagonal structure, $\Phi_{\overline{ph}}$ adds a diagonal structure and $\Phi_{pp}$ adds a constant background and a "cross"-like structure. Note that the description in Ref. [24] only refers to the SIAM model and considers solely the one-band case with local quantities and a constant local interaction $U$. An extension to a multi-orbital formalism and momentum-dependent vertices and interaction $U$ follows naturally, see Ref. [28] for a purely local description, and will be discussed hereafter.

[14] In Ref. [24] the local vertex functions were calculated for a value of $U = t$.

The simplest quantity to describe in terms of their asymptotics is the fully irreducible vertex $\Lambda^{qkk'}_{1234}$, for which

$$\Lambda^{\mathrm{asympt;qkk'}}_{1234;\sigma\sigma'} \approx \mathcal{U}^{\mathbf{qkk'}}_{1234;\sigma\sigma'}. \tag{1.65}$$

Constructing d, m, s and t combinations of the full non-local interaction in Eq. (??) is useful for future calculations,

$$\begin{aligned}
\mathcal{U}^{\mathbf{qkk'}}_{\mathrm{d};1234} &= 2\mathcal{U}^{\mathbf{q}}_{1234} - \tilde{\mathcal{U}}^{\mathbf{k'-k}}_{1234}, && (1.66a)\\
\mathcal{U}^{\mathbf{qkk'}}_{\mathrm{m};1234} &= -\tilde{\mathcal{U}}^{\mathbf{k'-k}}_{1234}, && (1.66b)\\
\mathcal{U}^{\mathbf{qkk'}}_{\mathrm{s};1234} &= \mathcal{U}^{\mathbf{k+k'-q}}_{1234} + \tilde{\mathcal{U}}^{\mathbf{k'-k}}_{1234} \qquad \text{and} && (1.66c)\\
\mathcal{U}^{\mathbf{qkk'}}_{\mathrm{t};1234} &= \mathcal{U}^{\mathbf{k+k'-q}}_{1234} - \tilde{\mathcal{U}}^{\mathbf{k'-k}}_{1234}. && (1.66d)
\end{aligned}$$

Next, we begin examining the diagrams contributing to $\Phi_r$. A smart choice introduced in Ref. [24] is to classify these diagrams in each channel further into three categories:

- Class 1: These are diagrams which connect incoming and outgoing particles only by a single interaction vertex. These diagrams are bubble diagrams and hence only depend on a single bosonic frequency and momentum. The sum of all these diagrams will be encoded in the variable $\mathcal{K}^{(1);q}_{r;1234;\sigma\sigma'}$, see the first line in Fig. ??.
- Class 2: Diagrams corresponding to class 2 are identified by them having either the incoming *or* outgoing particles connected to the same interaction vertex. These diagrams depend on the bosonic transfer frequency and momentum and one fermionic frequency and momentum. The sum of all these diagrams will be encoded in the variables $\mathcal{K}^{(2);qk}_{r;1234;\sigma\sigma'}$ and $\overline{\mathcal{K}}^{(2);qk}_{r;1234;\sigma\sigma'}$, depending on whether the outgoing particles ($\mathcal{K}$) or the incoming particles ($\overline{\mathcal{K}}$) are connected to the interaction vertex, see the second line in Fig. ??. $\mathcal{K}$ and $\overline{\mathcal{K}}$ are related to each other via time-reversal symmetry, where in the case of a system with additional SU(2)-symmetry corresponds to [15] $\mathcal{K}^{(2);qk}_{r;1234;\sigma\sigma'} = \overline{\mathcal{K}}^{(2);qk}_{r;1234;\sigma\sigma'}$.
- Class 3: These are diagrams where each external Green's function is connected to a different interaction vertex. These diagrams are parametrized by three independent frequency and momentum arguments and are encoded in the "rest" function $\mathcal{R}^{qkk'}_{r;1234;\sigma\sigma'}$.

In the following, we will refer to $\mathcal{K}^{(1)}$, $\mathcal{K}^{(2)}$ and $\overline{\mathcal{K}}^{(2)}$ as kernel functions. One now can — with the help of these classifications — write the reducible $\Phi_r$-vertices as a decomposition of the different kernel functions and the "rest"-function

$$\Phi^{qkk'}_{r;1234;\sigma\sigma'} = \mathcal{K}^{(1);q}_{r;1234;\sigma\sigma'} + \mathcal{K}^{(2);qk}_{r;1234;\sigma\sigma'} + \mathcal{K}^{(2);qk'}_{r;1234;\sigma\sigma'} + \mathcal{R}^{qkk'}_{r;1234;\sigma\sigma'}. \tag{1.67}$$

This composition is *per se* exact. However, to take advantage of this formalism for numerical calculations, one discards the "rest"-function due to its additional frequency dependence entering the inner propagators, resulting in a quicker drop-off in all frequency directions. Thus, the approximation

$$\Phi^{\mathrm{asympt};qkk'}_{r;1234;\sigma\sigma'} \approx \mathcal{K}^{(1);q}_{r;1234;\sigma\sigma'} + \mathcal{K}^{(2);qk}_{r;1234;\sigma\sigma'} + \mathcal{K}^{(2);qk'}_{r;1234;\sigma\sigma'} \tag{1.68}$$

serves as a valid expression for the high-frequency regime of $\Phi_r$ and becomes exact for $\nu, \nu' \to \infty$ [24]. As we can see above, the quantities on the right hand side of Eq. (??) only depend on one

(q) or two (q, k) Matsubara frequencies and momenta instead of three, reducing the numerical complexity and memory usage for $\Phi_r$ significantly. Using these explicit asymptotics results in a tremendous improvement of the convergence of numerical results over a simple truncation in $\Phi_r$ for high frequencies [24]. The explicit expressions for the kernel functions needed to calculate the reducible diagrams can be obtained by [10, 30, 31]

$$\mathcal{K}^{(1);\mathrm{q}}_{\mathrm{ph};1234;\sigma\sigma'} = \sum_{\substack{abcd;k_1k_2\\ \sigma_1\sigma_2}} \mathcal{U}^{\mathbf{qk}_1}_{12ab;\sigma\sigma_1} \chi^{\mathrm{qk}_1\mathrm{k}_2}_{badc;\sigma_1\sigma_2} \mathcal{U}^{\mathbf{q}(-\mathbf{k}_2)}_{cd34;\sigma_2\sigma'}, \tag{1.69}$$

$$\mathcal{K}^{(1);\mathrm{q}}_{\mathrm{pp};1234;\sigma\sigma'} = \frac{1}{4} \sum_{\substack{abcd;k_1k_2\\ \sigma_1\sigma_2\sigma_3\sigma_4}} \mathcal{U}^{\mathbf{qk}_1}_{\substack{1b3a\\ \sigma\sigma_2\sigma'\sigma_1}} \chi^{\mathrm{qk}_1\mathrm{k}_2}_{\substack{acbd\\ \sigma_1\sigma_3\sigma_2\sigma_4}} \mathcal{U}^{\mathbf{q}(-\mathbf{k}_2)}_{\substack{d2c4\\ \sigma_4\sigma\sigma_3\sigma'}}, \tag{1.70}$$

$$\begin{aligned}\mathcal{K}^{(2);\mathrm{qk}}_{\mathrm{ph};1234;\sigma\sigma'} &= \sum_{\substack{ab;k_1\\ \sigma_1}} F^{\mathrm{qkk}_1}_{12ab;\sigma\sigma_1} \chi^{\mathrm{qk}_1\mathrm{k}_1}_{0;baab} \mathcal{U}^{\mathbf{q}(-\mathbf{k}_1)}_{ba34;\sigma_1\sigma'} - \mathcal{K}^{(1);\mathrm{q}}_{\mathrm{ph};1234;\sigma\sigma'} \\ &= - \sum_{\substack{abcd;k_1\\ \sigma_1}} \left(\chi^{\mathrm{qk}}_{0;12dc}\right)^{-1} \chi^{\mathrm{qkk}_1}_{cdab;\sigma\sigma_1} \mathcal{U}^{\mathbf{q}(-\mathbf{k}_1)}_{ba34;\sigma_1\sigma'} - \mathcal{K}^{(1);\mathrm{q}}_{\mathrm{ph};1234;\sigma\sigma'}\end{aligned} \tag{1.71}$$

$$\begin{aligned}\mathcal{K}^{(2);\mathrm{qk}}_{\mathrm{pp};1234;\sigma\sigma'} &= \frac{1}{2} \sum_{\substack{ab;k_1\\ \sigma_1\sigma_2\sigma_3\sigma_4}} F^{\mathrm{qkk}_1}_{\substack{\mathrm{pp};1a3b\\ \sigma\sigma_1\sigma'\sigma_2}} \chi^{\mathrm{qk}_1\mathrm{k}_1}_{0;baab} \mathcal{U}^{\mathbf{q}(-\mathbf{k}_1)}_{\substack{b3a4\\ \sigma_2\sigma\sigma_1\sigma'}} - \mathcal{K}^{(1);\mathrm{q}}_{\mathrm{pp};1234;\sigma\sigma'} \\ &= -\frac{1}{2} \sum_{\substack{abcd;k_1\\ \sigma_1\sigma_2\sigma_3\sigma_4}} \left(\chi^{\mathrm{pp};\mathrm{qk}}_{0;1d2c}\right)^{-1} \chi^{\mathrm{qkk}_1}_{\substack{\mathrm{pp};cadb\\ \sigma\sigma_1\sigma'\sigma_2}} \mathcal{U}^{\mathbf{q}(-\mathbf{k}_1)}_{\substack{b3a4\\ \sigma_2\sigma\sigma_1\sigma'}} - \mathcal{K}^{(1);\mathrm{q}}_{\mathrm{pp};1234;\sigma\sigma'}\end{aligned} \tag{1.72}$$

where the subtraction of $\mathcal{K}^{(1)}$ in the kernel-2 function is to account for double-counting of diagrams and $\mathcal{U}^{\mathbf{qk}'}_{1234;\sigma\sigma'}$ is obtained from Eq. (**??**) by setting $\mathbf{k} = \mathbf{0}$. Both kernel functions are displayed diagrammatically for the ph-channel in Fig. **??**. The density and magnetic contributions to the

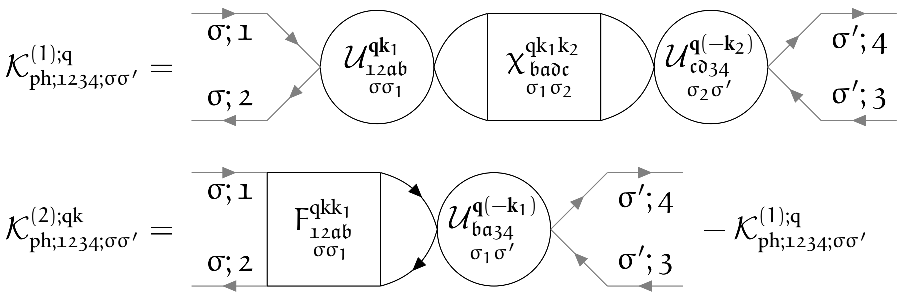



**FIGURE 1.16** – Diagrammatic version of the kernel-1 and kernel-2 functions of Eq. (**??**) and Eq. (**??**).

kernel functions read[15]

$$\mathcal{K}^{(1);\mathrm{q}}_{d;1234} = \sum_{\substack{abcd\\ k_1k_2}} \mathcal{U}^{\mathbf{qk}_1}_{d;12ab} \chi^{\mathrm{qk}_1\mathrm{k}_2}_{d;badc} \mathcal{U}^{\mathbf{q}(-\mathbf{k}_2)}_{d;cd34}, \tag{1.73a}$$

$$\mathcal{K}^{(1);\mathrm{q}}_{m;1234} = \sum_{\substack{abcd\\ k_1k_2}} \mathcal{U}^{\mathbf{qk}_1}_{m;12ab} \chi^{\mathrm{qk}_1\mathrm{k}_2}_{m;badc} \mathcal{U}^{\mathbf{q}(-\mathbf{k}_2)}_{m;cd34}, \tag{1.73b}$$

[15] Remember, that for density and magnetic contributions, frequencies should be in ph notation whereas for (s)inglet and (t)riplet contributions, frequencies should be in pp notation.

$$\begin{aligned}\mathcal{K}^{(2);qk}_{d;1234} &= \sum_{ab;k_1} F^{qkk_1}_{d;12ab}\,\chi^{qk_1k_1}_{0;baab}\,\mathcal{U}^{\mathbf{q}(-\mathbf{k}_1)}_{d;ba34} - \mathcal{K}^{(1);q}_{d;1234} \\ &= -\sum_{abcd;k_1} \left(\chi^{qk}_{0;12dc}\right)^{-1} \chi^{qkk_1}_{d;cdab}\,\mathcal{U}^{\mathbf{q}(-\mathbf{k}_1)}_{d;ba34} - \mathcal{K}^{(1);q}_{d;1234} \qquad \text{and}\end{aligned} \tag{1.73c}$$

$$\begin{aligned}\mathcal{K}^{(2);qk}_{m;1234} &= \sum_{ab;k_1} F^{qkk_1}_{m;12ab}\,\chi^{qk_1k_1}_{0;baab}\,\mathcal{U}^{\mathbf{q}(-\mathbf{k}_1)}_{m;ba34} - \mathcal{K}^{(1);q}_{m;1234} \\ &= -\sum_{abcd;k_1} \left(\chi^{qk}_{0;12dc}\right)^{-1} \chi^{qkk_1}_{m;cdab}\,\mathcal{U}^{\mathbf{q}(-\mathbf{k}_1)}_{m;ba34} - \mathcal{K}^{(1);q}_{m;1234}.\end{aligned} \tag{1.73d}$$

The s/t combinations can be constructed analogously for the pp-channel (written in pp frequency notation):

$$\mathcal{K}^{(1);q}_{s;1234} = \frac{1}{4}\sum_{\substack{abcd\\k_1k_2}} \mathcal{U}^{\mathbf{qk}_1}_{s;1b3a}\chi^{qk_1k_2}_{s;acbd}\mathcal{U}^{\mathbf{q}(-\mathbf{k}_2)}_{s;d2c4} \overset{(??)}{=} \frac{1}{2}\sum_{\substack{abcd\\k_1k_2}} \mathcal{U}^{\mathbf{qk}_1}_{s;1b3a}\chi^{pp;qk_1k_2}_{acbd;\uparrow\downarrow}\mathcal{U}^{\mathbf{q}(-\mathbf{k}_2)}_{s;d2c4}, \tag{1.74a}$$

$$\mathcal{K}^{(1);q}_{t;1234} = \frac{1}{4}\sum_{\substack{abcd\\k_1k_2}} \mathcal{U}^{\mathbf{qk}_1}_{t;1b3a}\chi^{qk_1k_2}_{t;acbd}\mathcal{U}^{\mathbf{q}(-\mathbf{k}_2)}_{t;d2c4} \overset{(??)}{=} 0, \tag{1.74b}$$

$$\begin{aligned}\mathcal{K}^{(2);qk}_{s;1234} &= \frac{1}{2}\sum_{ab;k_1} F^{qkk_1}_{s;1a3b}\,\chi^{qk_1k_1}_{0;baab}\,\mathcal{U}^{\mathbf{q}(-\mathbf{k}_1)}_{s;b3a4} - \mathcal{K}^{(1);q}_{s;1234} \\ &= -\frac{1}{2}\sum_{abcd;k_1} \left(\chi^{pp;qk}_{0;1d2c}\right)^{-1} \chi^{qkk_1}_{s;cadb}\,\mathcal{U}^{\mathbf{q}(-\mathbf{k}_1)}_{s;b3a4} - \mathcal{K}^{(1);q}_{s;1234} \\ &\overset{(??)}{=} -\sum_{abcd;k_1} \left(\chi^{pp;qk}_{0;1d2c}\right)^{-1} \chi^{pp;qkk_1}_{cadb;\uparrow\downarrow}\,\mathcal{U}^{\mathbf{q}(-\mathbf{k}_1)}_{s;b3a4} - \mathcal{K}^{(1);q}_{s;1234} \qquad \text{and}\end{aligned} \tag{1.74c}$$

$$\begin{aligned}\mathcal{K}^{(2);qk}_{t;1234} &= \frac{1}{2}\sum_{ab;k_1} F^{qkk_1}_{t;1a3b}\,\chi^{qk_1k_1}_{0;baab}\,\mathcal{U}^{\mathbf{q}(-\mathbf{k}_1)}_{t;b3a4} - \mathcal{K}^{(1);q}_{t;1234} \\ &= -\frac{1}{2}\sum_{abcd;k_1} \left(\chi^{pp;qk}_{0;1d2c}\right)^{-1} \chi^{qkk_1}_{t;cadb}\,\mathcal{U}^{\mathbf{q}(-\mathbf{k}_1)}_{t;b3a4} - \mathcal{K}^{(1);q}_{t;1234} \overset{(??)}{=} 0.\end{aligned} \tag{1.74d}$$

Finally, the asymptotic part of the full vertex $\left(F^{\text{asympt};qkk'}_{r;1234;\sigma\sigma'}\right)$ can now be assembled using the asymptotic contributions of $\Lambda^{\text{asympt};qkk'}_{1234;\sigma\sigma'}$ and $\Phi^{\text{asympt};qkk'}_{r;1234;\sigma\sigma'}$ through the parquet decomposition of Eq. (**??**). This results in

$$F^{\text{asympt};qkk'}_{1234;\sigma\sigma'} - \mathcal{U}^{\mathbf{qkk}'}_{1234;\sigma\sigma'} = \Phi^{\text{asympt};qkk'}_{ph;1234;\sigma\sigma'} + \Phi^{\text{asympt};qkk'}_{\overline{ph};1234;\sigma\sigma'} + \Phi^{\text{asympt};qkk'}_{pp;1234;\sigma\sigma'}, \tag{1.75}$$

or explicitly with kernel-1 and kernel-2 functions,

$$\begin{aligned}F^{\text{asympt};qkk'}_{1234;\sigma\sigma'} - \mathcal{U}^{\mathbf{qkk}'}_{1234;\sigma\sigma'} = {}& \mathcal{K}^{(1);q}_{ph;1234;\sigma\sigma'} + \mathcal{K}^{(2);qk}_{ph;1234;\sigma\sigma'} + \mathcal{K}^{(2);qk'}_{ph;1234;\sigma\sigma'} \\ & + \mathcal{K}^{(1);q}_{\overline{ph};1234;\sigma\sigma'} + \mathcal{K}^{(2);qk}_{\overline{ph};1234;\sigma\sigma'} + \mathcal{K}^{(2);qk'}_{\overline{ph};1234;\sigma\sigma'} \\ & + \mathcal{K}^{(1);q}_{pp;1234;\sigma\sigma'} + \mathcal{K}^{(2);qk}_{pp;1234;\sigma\sigma'} + \mathcal{K}^{(2);qk'}_{pp;1234;\sigma\sigma'}.\end{aligned} \tag{1.76}$$

The asymptotic behavior of the irreducible vertex in channel r follows from Eq. (**??**),

$$\Gamma^{\text{asympt};qkk'}_{r;1234;\sigma\sigma'} = F^{\text{asympt};qkk'}_{1234;\sigma\sigma'} - \Phi^{\text{asympt};qkk'}_{r;1234;\sigma\sigma'} \tag{1.77}$$

and reads for the ph-channel explicitly

$$\begin{aligned}\Gamma^{\text{asympt;qkk}'}_{\text{ph;1234;}\sigma\sigma'} &= \mathcal{U}^{\mathbf{qkk}'}_{1234;\sigma\sigma'} + \Phi^{\text{asympt;qkk}'}_{\overline{\text{ph}};1234;\sigma\sigma'} + \Phi^{\text{asympt;qkk}'}_{\text{pp;1234;}\sigma\sigma'} \\ &= \mathcal{U}^{\mathbf{qkk}'}_{1234;\sigma\sigma'} - \Phi^{\text{asympt;}(k'-k)(k'-q)k'}_{\text{ph;3214;}\overline{\sigma'\sigma}} + \Phi^{\text{asympt;}(k+k'-q)kk'}_{\text{pp;1234;}\sigma\sigma'},\end{aligned} \tag{1.78}$$

where we used Eq. (**??**) to transform the $\overline{\text{ph}}$-channel reducible vertex to a ph-channel reducible one. From this follows the irreducible vertex in the density and magnetic channel (in ph frequency notation) [25]

$$\begin{aligned}\Gamma^{\text{asympt;qkk}'}_{\text{d;1234}} = \mathcal{U}^{\mathbf{qkk}'}_{\text{d;1234}} &- \frac{1}{2}\Phi^{\text{asympt;}(k'-k)(k'-q)k'}_{\text{d;3214}} - \frac{3}{2}\Phi^{\text{asympt;}(k'-k)(k'-q)k'}_{\text{m;3214}} \\ &+ \frac{1}{2}\Phi^{\text{asympt;}(k+k'-q)kk'}_{\text{s;1234}} + \frac{3}{2}\Phi^{\text{asympt;}(k+k'-q)kk'}_{\text{t;1234}},\end{aligned} \tag{1.79a}$$

$$\begin{aligned}\Gamma^{\text{asympt;qkk}'}_{\text{m;1234}} = \mathcal{U}^{\mathbf{qkk}'}_{\text{m;1234}} &- \frac{1}{2}\Phi^{\text{asympt;}(k'-k)(k'-q)k'}_{\text{d;3214}} + \frac{1}{2}\Phi^{\text{asympt;}(k'-k)(k'-q)k'}_{\text{m;3214}} \\ &- \frac{1}{2}\Phi^{\text{asympt;}(k+k'-q)kk'}_{\text{s;1234}} + \frac{1}{2}\Phi^{\text{asympt;}(k+k'-q)kk'}_{\text{t;1234}},\end{aligned} \tag{1.79b}$$

where the $\Phi^{\text{asympt;qkk}'}_{\text{r;1234}}$ are given by the kernel functions of Eq. (**??**) and Eq. (**??**). The irreducible vertices in channels $\overline{\text{ph}}$ and pp in m/d and s/t combinations can be constructed analogously.

### 1.7.2 $\mathcal{U}$-range asymptotics

Besides the kernel-function formalism introduced above, there also exist many other asymptotic schemes for vertex functions. The second one we would like to cover in this thesis is the "$\mathcal{U}$-range" approach, where the irreducible vertex $\Gamma^{\text{qkk}'}_{1234;\sigma\sigma'}$ is padded with the bare non-local Coulomb interaction $\mathcal{U}^{\mathbf{qkk}'}_{1234;\sigma\sigma'}$, see Eq. (**??**), in the high-frequency regime. This in turn means that there is no complicated frequency structure for the asymptotic region. We denote some equations that are necessary to employ these types of asymptotics and follow Ref. [29] with the addition of multi-orbital notation. First, we will split up the frequencies into a "core" and an "asymptotic" region and denote quantities that exist in those frequency regimes with an additional sub- or superscript $\in$ {core, asympt}. Generally speaking, we require $n_{\text{asympt}} \geq n_{\text{core}}$, where $n$ is the number of fermionic Matsubara frequencies. Furthermore, if a quantity is written as $(A^{\text{asympt}})_{\text{core}}$ it means that the asymptotic value of $A$ is cut to the core frequency region. Additionally, the size change and matrix inverse do *not* commute, i.e. $A^{-1}_{\text{core}} \equiv (A^{\text{core}})^{-1} \neq (A^{-1})_{\text{core}}$. This only commutes if the matrix $A$ is block-diagonal.

The advantages of this method are not only the simpler way of treating asymptotics, as no complicated kernel-functions are needed, but also considers the ease of extending $n_{\text{asympt}}$ without too much computational effort. As we will see shortly, we only need to solve the $n_{\text{asympt}}$-size of the local inverse BSE once and the rest of the equations can be performed within $n_{\text{core}}$. This allows us to increase $n_{\text{asympt}}$ to very large sizes without sacrificing too much computational time and memory. Fortunately, as will be seen later in Sec. **??**, this only needs to be done once at the beginning of the routine of the dynamical vertex approximation, since the local irreducible vertex is constant throughout the D$\Gamma$A algorithm. Only the bare bubble susceptibility $\chi^{\text{qk}}_{0;1234}$ has to be known additionally, however, due to the simple frequency dependence of this quantity, it is easily available for $n_{\text{asympt}}$ Matsubara frequencies.

We begin by extracting the local irreducible vertex $\Gamma^{\omega\nu\nu'}_{1234,\sigma\sigma'}$ from the generalized susceptibility and the bare bubble susceptibility obtained from the impurity solver in the core region,

$$\Gamma^{\text{core};\omega\nu\nu'}_{1234;\sigma\sigma'} = \left(\chi^{\text{core};\omega\nu\nu'}_{1234;\sigma\sigma'}\right)^{-1} - \left(\chi^{\text{core};\omega\nu\nu'}_{0;1234}\right)^{-1}. \tag{1.80}$$

In order to incorporate the asymptotic bare (and local) Coulomb interaction $\mathcal{U}^{\text{asympt}}_{1234;\sigma\sigma'}$ into the local irreducible vertex, we need to solve the $n_{\text{asympt}}$-size BSE,

$$\delta\Gamma^{\text{core};\omega\nu\nu'}_{1234;\sigma\sigma'} + \mathcal{U}^{\text{asympt}}_{1234;\sigma\sigma'} = \left(\chi^{\text{asympt};\omega\nu\nu'}_{1234;\sigma\sigma'}\right)^{-1} - \left(\chi^{\text{asympt};\omega\nu\nu'}_{0;1234}\right)^{-1}, \tag{1.81}$$

where $\delta\Gamma^{\text{core};\omega\nu\nu'}_{1234;\sigma\sigma'}$ is the irreducible vertex minus the bare Coulomb interaction. Additionally, $\chi^{\text{asympt};\omega\nu\nu'}_{1234;\sigma\sigma'}$ must equal the generalized susceptibility obtained from the impurity solver in the $n_{\text{core}}$ region. Since we do not know how the generalized susceptibility behaves outside the core frequency region at this stage, this approach does not look promising. However, since we have fixed the irreducible vertex to the bare Coulomb interaction for high frequencies, we can obtain the local (corrected) irreducible vertex with the help of an intermediary susceptibility, where

$$\left(\tilde{\chi}^{\text{asympt};\omega\nu\nu'}_{1234;\sigma\sigma'}\right)^{-1} = \left(\chi^{\text{asympt};\omega\nu\nu'}_{0;1234}\right)^{-1} + \mathcal{U}^{\text{asympt}}_{1234;\sigma\sigma'}. \tag{1.82}$$

From the above two equations we can formulate the BSE as

$$\delta\Gamma^{\text{core};\omega\nu\nu'}_{1234;\sigma\sigma'} = \left(\chi^{\text{asympt};\omega\nu\nu'}_{1234;\sigma\sigma'}\right)^{-1}_{\text{core}} - \left(\tilde{\chi}^{\text{asympt};\omega\nu\nu'}_{1234;\sigma\sigma'}\right)^{-1}_{\text{core}}, \tag{1.83}$$

where after some rewriting and algebra [29]

$$\Gamma^{\text{core};\omega\nu\nu'}_{1234;\sigma\sigma'} + \mathcal{U}^{\text{core}}_{1234;\sigma\sigma'} = \left(\chi^{\text{core};\omega\nu\nu'}_{1234;\sigma\sigma'}\right)^{-1} - \left[\left(\tilde{\chi}^{\text{asympt};\omega\nu\nu'}_{1234;\sigma\sigma'}\right)_{\text{core}}\right]^{-1}. \tag{1.84}$$

We now have rewritten the irreducible vertex in terms of quantities that are already available from the impurity solver, such as $\chi^{\text{core};\omega\nu\nu'}_{1234;\sigma\sigma'}$, and the intermediate susceptibility, which is easily retrieved from the $n_{\text{asympt}}$-range bare bubble susceptibility and the bare Coulomb interaction. This can be seen as an asymptotic correction to the $n_{\text{core}}$-size BSE. Due to the frequency structure of $\tilde{\chi}^{\text{asympt};\omega\nu\nu'}_{1234;\sigma\sigma'}$, this does not require a lot of computational effort or memory and hence is a very resource efficient approach.

The extension to non-local asymptotics is straightforward and can be derived in a few steps, where we will only outline the most important results. For a more in-depth discussion, we refer the reader to Ref. [29]. At the beginning, we define

$$\Gamma^{X;\omega\nu\nu'}_{1234;\sigma\sigma'} = \left(\chi^{X;\omega\nu\nu'}_{1234;\sigma\sigma'}\right)^{-1} - \left(\chi^{X;\omega\nu\nu'}_{0;1234}\right)^{-1} \quad \text{and} \tag{1.85}$$

$$\left(\chi^{*;X;\omega\nu\nu'}_{1234;\sigma\sigma'}\right)^{-1} = \left(\chi^{X;\omega\nu\nu'}_{1234;\sigma\sigma'}\right)^{-1} - \mathcal{U}^{X}_{1234;\sigma\sigma'}, \tag{1.86}$$

where $X \in \{\text{core}, \text{asympt}\}$ and $\chi^{*;X;\omega\nu\nu'}_{1234;\sigma\sigma'}$ is the same auxiliary susceptibility as already seen in Eq. (**??**). Both of these equations have to be satisfied for each frequency range $n_X$. From the lower equation we obtain

$$\chi^{*;X;qkk'}_{1234;\sigma\sigma'} = \chi^{X;qkk'}_{1234;\sigma\sigma'} + \sum_{\substack{k_1k_2;\mathfrak{abcdef} \\ \sigma_1\sigma_2\sigma_3\sigma_4}} \chi^{X;qkk_1}_{12\mathfrak{ba};\sigma\sigma_1} \mathcal{U}^{X;\mathbf{qk}_1\mathbf{k}_2}_{\mathfrak{abcd};\sigma_1\sigma_2} \chi^{X;qk_2k'}_{\mathfrak{dcef};\sigma_2\sigma_3} \left(\mathbb{1}_{\mathfrak{fe}34} - \mathcal{U}^{X;\mathbf{qk}_1\mathbf{k}_2}_{\mathfrak{fehg};\sigma_3\sigma_4} \chi^{X;qk_2k_1}_{\mathfrak{gh}34;\sigma_4\sigma'}\right)^{-1}. \tag{1.87}$$

From this one can arrive at an expression for the generalized physical susceptibility [29]

$$\chi^{\mathrm{core;q}}_{1234;\sigma\sigma'} = \sum_{\mathrm{kk}'} \chi^{\mathrm{core;qkk}'}_{1234;\sigma\sigma'} = \sum_{\mathrm{kk}'} \left[ \left( \chi^{*;\mathrm{core;qkk}'}_{1234;\sigma\sigma'} + \chi^{\mathrm{asympt;qkk}'}_{0;1234;\sigma\sigma'} - \chi^{\mathrm{core;qkk}'}_{0;1234;\sigma\sigma'} \right)^{-1} + \mathcal{U}^{\mathrm{core;\mathbf{qkk}}'}_{1234;\sigma\sigma'} \right]^{-1}. \tag{1.88}$$

This is especially useful, since the generalized physical susceptibility enters the SDE (**??**) with the three-leg vertices of Eq. (**??**). Hence, this scheme proves to be particularly powerful [29, 32] for this explicit form of the equation of motion of $\Sigma^{\mathrm{k}}_{12}$. Lastly, it can be shown [33], that for the three-leg vertex $\gamma^{\mathrm{qk}}_{\mathrm{d/m;1234}} \xrightarrow{\nu\to\infty} \mathbb{1}_{1234}$ holds.

### 1.7.3 High-frequency Bethe-Salpeter equation with vertex asymptotics

Sometimes it is convenient to separate low- and high-frequency components of a quantity to simplify calculations. Some DMFT solvers [34] employ such a frequency separation to calculate the self-energy: a high-frequency asymptotic expansion derived from the spectral function's lowest moments is incorporated in the numerical solution of the Dyson equation at low frequencies. Inspired by this idea, we can proceed similarly for the irreducible vertex $\Gamma$, working with the framework of the Bethe-Salpeter equation instead of the Dyson equation. By explicitly using the SU(2)-symmetric combinations d, m — in which the BSE for the ph-channel becomes diagonal[16] — we are able to reduce the computational effort by using a high-frequency expansion of the vertex functions. Following [23, 25, 27], the vertex in the low-frequency regime can be obtained by the regular BSE, where an additional high-frequency correction term is added. In the spirit of the end of Sec. **??**, the BSE of Eq. (**??**) can be written as a matrix equation (in grouped indices nm) for spin combinations d, m in the following fashion

$$\boldsymbol{\chi}^{\mathrm{q}} = \boldsymbol{\chi}^{\mathrm{q}}_0 - \frac{1}{\beta^2} \boldsymbol{\chi}^{\mathrm{q}}_0 \boldsymbol{\Gamma}^{\mathrm{q}} \boldsymbol{\chi}^{\mathrm{q}}. \tag{1.89}$$

Similar to Refs. [23, 25, 27], we will now split the matrices $\boldsymbol{F}$, $\boldsymbol{\Gamma}$ and $\boldsymbol{\chi}$ in segments of low and high frequencies. We will denote this by an additional superscript $(\mathrm{L/H}, \mathrm{L/H})$, where $(\mathrm{L}, \mathrm{L})$ denotes the segment, where $\nu$ and $\nu'$ are in the low frequency region and, e.g. $(\mathrm{L}, \mathrm{H})$ denotes the segment, where $\nu$ is in the low- and $\nu'$ in the high-frequency region. We will therefore construct a set of low frequencies $\mathcal{I}_{\mathrm{L}}$ and a set of high frequencies $\mathcal{I}_{\mathrm{H}}$. This means, that for $\boldsymbol{\Gamma}^{\mathrm{LH}}$, $\nu \in \mathcal{I}_{\mathrm{L}}$ and $\nu' \in \mathcal{I}_{\mathrm{H}}$. In the LH, HL and HH blocks, where atleast one frequency is considered large, we will employ explicit asymptotics for the vertex functions, as already discussed in Sec. **??**. The goal is to calculate the vertex function with the BSE for the low-frequency region where we will expect a correction term arising from the low-frequency cutoff which employs the use of high-frequency blocks of the BSE components. The size of $\mathcal{I}_{\mathrm{L}}$ has to be atleast the number of bosonic Matsubara frequencies $\omega$ [25] and might differ if non-local correlations are included [35]. Regarding $\mathcal{I}_{\mathrm{H}}$, there is no minimum size and the maximum size of the high-frequency region is only bounded by the availability of computational resources and memory.
Using this idea, we will solve Eq. (**??**) for $\boldsymbol{\Gamma}$ as

$$\begin{pmatrix} \boldsymbol{\Gamma}^{\mathrm{LL}} & \boldsymbol{\Gamma}^{\mathrm{LH}} \\ \boldsymbol{\Gamma}^{\mathrm{HL}} & \boldsymbol{\Gamma}^{\mathrm{HH}} \end{pmatrix} = \beta^2 \left[ \begin{pmatrix} \boldsymbol{\chi}^{\mathrm{LL}} & \boldsymbol{\chi}^{\mathrm{LH}} \\ \boldsymbol{\chi}^{\mathrm{HL}} & \boldsymbol{\chi}^{\mathrm{HH}} \end{pmatrix}^{-1} - \begin{pmatrix} \boldsymbol{\chi}^{\mathrm{LL}}_0 & 0 \\ 0 & \boldsymbol{\chi}^{\mathrm{HH}}_0 \end{pmatrix}^{-1} \right] \tag{1.90}$$

[16] The same can be done for the s, t combinations in the pp-channel, however only the ph-channel will be shown.

where we omitted frequency labels for brevity. Multiplying on the left by $\chi$ and solving this matrix-like equation for $\boldsymbol{\Gamma}^{\mathrm{LL}}$, we find

$$\boldsymbol{\Gamma}^{\mathrm{LL}} = \beta^2 \left[ \left(\boldsymbol{\chi}^{\mathrm{LL}}\right)^{-1} - \left(\boldsymbol{\chi}_0^{\mathrm{LL}}\right)^{-1} \right] - \left(\boldsymbol{\chi}^{\mathrm{LL}}\right)^{-1} \boldsymbol{\chi}^{\mathrm{LH}} \boldsymbol{\Gamma}^{\mathrm{HL}}, \tag{1.91}$$

with

$$\boldsymbol{\chi}^{\mathrm{LH}} = -\boldsymbol{\chi}^{\mathrm{LL}} \boldsymbol{\Gamma}^{\mathrm{LH}} \left[ \boldsymbol{\Gamma}^{\mathrm{HH}} + \beta^2 \left(\boldsymbol{\chi}_0^{\mathrm{HH}}\right)^{-1} \right]^{-1}. \tag{1.92}$$

Combining these two equations yields for $\boldsymbol{\Gamma}^{\mathrm{LL}}$

$$\boldsymbol{\Gamma}^{\mathrm{LL}} = \beta^2 \left[ \left(\boldsymbol{\chi}^{\mathrm{LL}}\right)^{-1} - \left(\boldsymbol{\chi}_0^{\mathrm{LL}}\right)^{-1} \right] + \boldsymbol{\Gamma}^{\mathrm{LH}} \left[ \boldsymbol{\Gamma}^{\mathrm{HH}} + \beta^2 \left(\boldsymbol{\chi}_0^{\mathrm{HH}}\right)^{-1} \right]^{-1} \boldsymbol{\Gamma}^{\mathrm{HL}}. \tag{1.93}$$

The last term in the sum of Eq. (**??**) can be thought of as a correction to the inversion of the BSE in the low-frequency subspace, which otherwise reads

$$\boldsymbol{\Gamma}^{\mathrm{LL}} = \beta^2 \left[ \left(\boldsymbol{\chi}^{\mathrm{LL}}\right)^{-1} - \left(\boldsymbol{\chi}_0^{\mathrm{LL}}\right)^{-1} \right]. \tag{1.94}$$

The correction term can be obtained at a significantly lower cost compared to the full vertex — since for the high-frequency regimes of $\boldsymbol{\Gamma}$ we use either (i) kernel-1 and kernel-2 functions, which depend on one or two momentum and frequency dimensions less than the full vertex (or the generalized susceptibility), see Sec. **??** or (ii) the even more resource efficient "U-range" approach of Sec. **??** — and is therefore easily available for much larger asymptotic frequency ranges. Other sets of equations for $\boldsymbol{\Gamma}^{\mathrm{LL}}$[17] can be constructed easily by manipulating the original BSE (**??**). These methods would require the calculation of the full vertex for a large frequency range[18], which is more storage-heavy than the use of kernel-1 and kernel-2 functions or a padding of the bare Coulomb interaction. Thus, the approach outlined in this thesis is more efficient and is going to be a more practical one to use for multi-orbital calculations.

Since we have now assembled all necessary tools to describe the properties of correlated electrons on a lattice, let us introduce the DΓA's purely local predecessor: the dynamical mean-field theory (DMFT).

[17] One can already use Eq. (**??**) instead of Eq. (**??**), cf. "Method 2" in Ref. [25].

[18] Retrieving the vertex functions in a high-frequency regime is *per se* not difficult; they can be obtained for arbitrary frequency ranges from the impurity solver of choice. However, the computing time increases drastically the larger the frequency box and thus this method is not preferred over the one mentioned above.